Aspirra — Project Proposal

Course: CSE454 — Advanced Software Engineering
Institution: Alamein International University
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Date: October 2026

Overview

Aspirra is an AI-powered platform that turns a student's academic and career goals into a realistic, editable plan. It connects goals, courses, assignments, exams, study schedules, habits, skills, projects, and career preparation in one place, adapts the plan when the student falls behind, and warns about risks early — always with the student's approval before anything changes.

Study, habits, and career growth, all in one place.

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Part A — Project Overview

The project title, description, problem statement, related work, and team information.

A1 — Title, Description, Problem & Related Work

Title & description

Aspirra: Your AI Planner for Academic Success and Career Growth

Every student has big goals, but turning them into daily action is the hard part. Aspirra is an AI-powered platform that makes it simple.

Share a goal like “reach a 3.5 GPA” or “land a software internship”, and Aspirra builds your roadmap of milestones, tasks, habits, and deadlines. You stay in control by editing the plan, and Aspirra fits it around your real schedule.

Life gets busy and plans slip. When yours does, Aspirra warns you early and recommends a way back on track, and nothing changes until you approve it. Along the way it tracks your skills, projects, and portfolio, so every study session moves you closer to your career.

Study, habits, and career growth, all in one place.

The problem

Students don’t fail from a lack of ability. They fail because nothing connects their goals to their daily actions.

Deadlines sit in a calendar, tasks in a notes app, grades on the university portal, and career plans only in the student’s head. So goals stay too big to act on, schedules ignore real available time, and plans break when a single task is missed. Nobody warns the student early, and study stays disconnected from the skills and projects a career requires. The result is missed deadlines, lower grades, stress, and abandoned goals.

What Aspirra solves

Aspirra turns a goal into a realistic, editable plan, adapts it when the student falls behind, warns about risks early, and links daily study to career skills, all in one platform.

Who suffers from it

University students are the people most affected, especially:

- **First-year students**, who are still learning to plan and are adjusting to the university workload.
- **Students with heavy workloads**, who juggle many courses, projects, and part-time work or activities.
- **Senior students**, who must balance coursework and graduation projects with portfolios and job applications.
- **Students who fall behind**, who have no easy way to recover once a plan breaks.

Instructors and academic advisors are affected indirectly, because they deal with late submissions and struggling students. They are not the main users of the system.

Main features

#	Feature	Description
1	Student Profile	Academic level, major, interests, skills, career interests, and availability.

#	Feature	Description
2	Academic Goal Management	Semester goals, course goals, GPA targets, assignment goals, and exam goals.
3	AI Goal Planner	Convert a natural-language goal into milestones, tasks, habits, deadlines, and priorities.
4	AI Study Plan Generator	Create personalized study plans and recommend adjustments based on progress and available time.
5	Course Management	Manage courses, grades, assignments, projects, quizzes, exams, and course progress.
6	Assignment & Exam Planner	Track deadlines, priorities, preparation schedules, and exam countdowns.
7	Career Roadmap	Select a target career and track required skills, learning steps, and career progress.
8	Skill Tracking	Track technical and soft skills, skill levels, learning progress, and career skill requirements.
9	Project & Portfolio Management	Track projects, connect them to skills, and maintain a portfolio.
10	Habit Management	Track study, exercise, reading, coding, and other habits, including streaks and goal connections.
11	Smart Scheduling	Organize tasks using deadlines, priorities, workload, availability, and dependencies.
12	Adaptive Planning	Detect when a student falls behind and recommend rescheduling or reprioritizing work.
13	AI Student Coach	Provide personalized recommendations using the student's goals, tasks, habits, and progress.
14	Academic Performance Analytics	Show grade trends, course performance, study consistency, assignment completion, and GPA progress.
15	Prediction & Early Warning	Estimate goal completion and identify potential academic or deadline risks.
16	Dashboard	Show today's tasks, upcoming deadlines, goal progress, habit streaks, academic performance, and career progress.
17	Notifications	Provide assignment, exam, study, missed-task, goal, and deadline reminders.
18	Study Groups / Accountability	Support shared goals, study groups, accountability partners, and shared progress.
19	Internship & Job Preparation	Track applications, interview preparation, CV/portfolio progress, and required skills.

#	Feature	Description
20	Admin Dashboard	Support student management, system statistics, usage analytics, and activity monitoring.

Related work

System	Description	Limitations
Shovel	A student planner where AI turns natural language into tasks, imports timetables, syncs calendars, checks if the schedule is realistic, and tracks study streaks	Plans only short-term study tasks. No long-term goal roadmap, no career or skill tracking, no study groups, no early-warning prediction. Paid subscription
MyStudyLife	A free planner for classes, assignments, and exams, with deadline reminders and a Pomodoro timer.	Mostly manual entry. No AI planning, no goal breakdown, no adaptive rescheduling, no career or skill features.
StudntX	An AI assistant that builds study plans around the student's schedule and tracks habits and streaks.	Focuses only on study plans and habits. No career roadmap, no skill or portfolio tracking, no analytics or early warning, no study groups.
Reclaim.ai	A web-based smart calendar that automatically blocks time for tasks and habits, and suggests a time-blocked day that the user approves or adjusts.	Built as a general calendar, not for students. No course, grade, or exam tracking, no goal-to-plan breakdown, no career or skill features, no study groups. Its scheduling follows set rules rather than understanding the student's goals.

Comparison table

Feature	Shovel	MyStudyLife	StudntX	Reclaim.ai	Aspirra
Natural-language input to tasks	Yes	No	Yes	No	Yes
Full goal roadmap (milestones, tasks, habits, deadlines)	No	No	Partial	No	Yes
Course, assignment, and exam tracking	Yes	Yes	Yes	No	Yes
AI study plan around availability	Yes	No	Yes	Partial	Yes
Schedule feasibility check	Yes	No	No	No	Yes
Adaptive replanning when behind (student approves)	No	No	No	Partial	Yes
Habit tracking with streaks	Yes	No	Yes	Partial	Yes
Deadline reminders	Yes	Yes	Yes	Yes	Yes
Career roadmap and skill tracking	No	No	No	No	Yes

Feature	Shovel	MyStudyLife	StudntX	Reclaim.ai	Aspirra
Project and portfolio management	No	No	No	No	Yes
Prediction and early warning	No	No	No	No	Yes
Study groups and accountability	No	No	No	No	Yes
Admin dashboard	No	No	No	No	Yes

A2 — Team Table

ID	Name	University email	GitHub	Services
23101359	Ahmed Mohamed Ahmed	ahmed.abdo.2024@aiu.edu.eg	AbooSalh	`student-service`, `academic-service`, `course-service`, `assessment-service`
23101506	Yomna Mohamed Fekry	yomna.hosney.2024@aiu.edu.eg	YomnaElmenawy	`goal-planner-service`, `study-planner-service`, `scheduling-service`, `adaptation-service`
23101371	Arwa Mohey Hamdy	arwa.hamdy.2024@aiu.edu.eg	ArwaMohey	`career-service`, `skill-service`, `portfolio-service`, `job-preparation-service`
23101484	Mostafa Khamis Abozead	mostafa.abozead.2024@aiu.edu.eg	Mostafa-abozead	`habit-service`, `analytics-service`, `prediction-service`, `ai-coach-service`
23102359	Muhammed Hany Hamdy	mohamed.abuaaliyah.2024@aiu.edu.eg	muhammedhanii	`dashboard-service`, `notification-service`, `study-group-service`,

ID	Name	University email	GitHub	Services
				`administration-service`

Part B — Requirements & Design

Business and customer requirements, functional and non-functional requirements, system decomposition, interfaces, traceability, and use cases.

B3 — Business & Customer Requirements

Business Requirements

ID	Business Requirement
BR-01	Reduce missed assignment and exam deadlines among active students by 30% within one semester of use.
BR-02	Increase the number of active students who complete their semester goals by 25% within one semester of use.
BR-03	Reduce the time students spend planning their study schedule by 50% within the first month of use.
BR-04	Reach 1,000 registered students within 6 months of launch.
BR-05	Increase the number of students who keep a study habit for at least 4 weeks in a row by 30% within one semester of use.
BR-06	Have 60% of registered students set a target career and list their skills within 6 months of launch.
BR-07	Reduce the time spent on student account support by 40% within one year of launch.
BR-08	Keep at least 50% of registered students active every month within 6 months of launch.

Customer Requirements

User type 1: First-year students

ID	Customer Requirement	Linked to
CR-01	First-year students need to plan their courses and workload without any planning experience.	BR-03

ID	Customer Requirement	Linked to
CR-02	First-year students need to know what is due and when across all their courses.	BR-01
CR-03	First-year students need to build good study habits from the start.	BR-05

User type 2: Students with heavy workloads

ID	Customer Requirement	Linked to
CR-04	Students with heavy workloads need to fit courses, projects and part-time work into one realistic schedule.	BR-03
CR-05	Students with heavy workloads need to know what to do first when many deadlines collide.	BR-01
CR-06	Students with heavy workloads need to know early when their workload is too much to finish.	BR-02

User type 3: Senior students

ID	Customer Requirement	Linked to
CR-07	Senior students need to balance their graduation project and coursework with job preparation.	BR-02
CR-08	Senior students need to show their skills and projects to employers in one place.	BR-06
CR-09	Senior students need to know which skills they are missing for their target career.	BR-06

User type 4: Students who fall behind

ID	Customer Requirement	Linked to
CR-10	Students who fall behind need to be warned early, before a deadline is missed.	BR-01
CR-11	Students who fall behind need a clear way to recover after their plan breaks.	BR-02
CR-12	Students who fall behind need to stay motivated after missing tasks, with support from friends or study partners.	BR-08

User type 5: Administrator

ID	Customer Requirement	Linked to
CR-13	Admins need to manage student accounts without asking the developers.	BR-07
CR-14	Admins need to see how many students use the platform and how they use it.	BR-04
CR-15	Admins need to find and fix student account problems quickly.	BR-07
CR-16	Admins need to see how many students keep using the platform over time.	BR-08

B4 — Functional Requirements

High-level functional requirements

ID	High-level FR (verb title)	Owner	Linked to
FR-01	Manage Student Profile	Ahmed Mohamed Ahmed	CR-01
FR-02	Manage Academic Goals	Ahmed Mohamed Ahmed	CR-01, CR-02
FR-03	Manage Courses	Ahmed Mohamed Ahmed	CR-01, CR-02, CR-04
FR-04	Manage Assignments and Exams	Ahmed Mohamed Ahmed	CR-02, CR-04
FR-05	Plan Goals with AI	Yomna Mohamed Fekry	CR-01, CR-04
FR-06	Generate Study Plans	Yomna Mohamed Fekry	CR-01, CR-04, CR-06
FR-07	Enable Smart Scheduling	Yomna Mohamed Fekry	CR-04, CR-05
FR-08	Enable Adaptive Planning	Yomna Mohamed Fekry	CR-04, CR-05, CR-06, CR-11
FR-09	Manage Career Roadmaps	Arwa Mohey Hamdy	CR-07, CR-09
FR-10	Track Skills	Arwa Mohey Hamdy	CR-08, CR-09
FR-11	Manage Projects and Portfolios	Arwa Mohey Hamdy	CR-08
FR-12	Manage Internship and Job Preparation	Arwa Mohey Hamdy	CR-07, CR-08

ID	High-level FR (verb title)	Owner	Linked to
FR-13	Manage Habits	Mostafa Khamis Abozead	CR-03
FR-14	Analyze Study Workload	Mostafa Khamis Abozead	CR-06
FR-15	Predict and Warn Early	Mostafa Khamis Abozead	CR-06, CR-10, CR-11
FR-16	Coach Students with AI	Mostafa Khamis Abozead	CR-06, CR-10, CR-11, CR-12
FR-17	Display Student Dashboard	Muhammed Hany Hamdy	CR-01, CR-02, CR-03, CR-06, CR-07
FR-18	Send Notifications	Muhammed Hany Hamdy	CR-02, CR-03, CR-04, CR-05, CR-10
FR-19	Manage Study Groups and Accountability	Muhammed Hany Hamdy	CR-12
FR-20	Manage Administration	Muhammed Hany Hamdy	CR-13, CR-14, CR-15, CR-16

Low-level functional requirements

FR-01 — Manage Student Profile

Owner: Ahmed Mohamed Ahmed | **Linked to:** CR-01

ID	Requirement	Owner	Linked to
FR-01.1	The system shall allow a student to create a profile with academic level, major, interests, skills, career interests, and availability.	Ahmed Mohamed Ahmed	CR-01
FR-01.2	The system shall allow a student to update profile information including availability and career interests.	Ahmed Mohamed Ahmed	CR-01
FR-01.3	The system shall allow a student to view their complete profile details.	Ahmed Mohamed Ahmed	CR-01
FR-01.4	The system shall validate required profile fields before saving profile data.	Ahmed Mohamed Ahmed	CR-01

FR-02 — Manage Academic Goals

Owner: Ahmed Mohamed Ahmed | **Linked to:** CR-01, CR-02

ID	Requirement	Owner	Linked to
FR-02.1	The system shall allow a student to create an academic goal with a title, description, target date, and priority.	Ahmed Mohamed Ahmed	CR-01, CR-02
FR-02.2	The system shall allow a student to update the details of an existing academic goal.	Ahmed Mohamed Ahmed	CR-01
FR-02.3	The system shall allow a student to update the target date of an existing academic goal.	Ahmed Mohamed Ahmed	CR-02
FR-02.4	The system shall allow a student to mark an academic goal as completed.	Ahmed Mohamed Ahmed	CR-02
FR-02.5	The system shall allow a student to archive an academic goal.	Ahmed Mohamed Ahmed	CR-01
FR-02.6	The system shall allow a student to delete an academic goal they created.	Ahmed Mohamed Ahmed	CR-01

FR-03 — Manage Courses

Owner: Ahmed Mohamed Ahmed | **Linked to:** CR-01, CR-02, CR-04

ID	Requirement	Owner	Linked to
FR-03.1	The system shall allow a student to add a course with course name, course code, instructor, and semester details.	Ahmed Mohamed Ahmed	CR-01
FR-03.2	The system shall allow a student to record the credits and the weekly workload of each course.	Ahmed Mohamed Ahmed	CR-01, CR-04
FR-03.3	The system shall allow a student to view the progress of a course, including its completed and outstanding assignments.	Ahmed Mohamed Ahmed	CR-02, CR-04
FR-03.4	The system shall allow a student to update course information.	Ahmed Mohamed Ahmed	CR-01
FR-03.5	The system shall allow a student to remove a course.	Ahmed Mohamed Ahmed	CR-01

FR-04 — Manage Assignments and Exams

Owner: Ahmed Mohamed Ahmed | **Linked to:** CR-02, CR-04

ID	Requirement	Owner	Linked to
FR-04.1	The system shall allow a student to add an assignment with title, course, due date, priority, and description.	Ahmed Mohamed Ahmed	CR-02
FR-04.2	The system shall allow a student to add an exam with title, course, exam date, time, and location.	Ahmed Mohamed Ahmed	CR-02
FR-04.3	The system shall allow a student to track assignment status as not started, in progress, or completed.	Ahmed Mohamed Ahmed	CR-02, CR-04
FR-04.4	The system shall allow a student to set preparation reminders for upcoming exams.	Ahmed Mohamed Ahmed	CR-02

FR-05 — Plan Goals with AI

Owner: Yomna Mohamed Fekry | **Linked to:** CR-01, CR-04

ID	Requirement	Owner	Linked to
FR-05.1	The system shall allow a student to input a goal in natural language for AI processing.	Yomna Mohamed Fekry	CR-01
FR-05.2	The system shall generate a structured goal roadmap with milestones, tasks, and deadlines from natural language input.	Yomna Mohamed Fekry	CR-01
FR-05.3	The system shall allow a student to review AI-generated milestones and tasks before saving.	Yomna Mohamed Fekry	CR-04
FR-05.4	The system shall allow a student to edit AI-generated milestones before saving.	Yomna Mohamed Fekry	CR-04
FR-05.5	The system shall allow a student to edit AI-generated tasks before saving.	Yomna Mohamed Fekry	CR-04
FR-05.6	The system shall require student approval before applying AI-generated goals to their plan.	Yomna Mohamed Fekry	CR-04

FR-06 — Generate Study Plans

Owner: Yomna Mohamed Fekry | **Linked to:** CR-01, CR-04, CR-06

ID	Requirement	Owner	Linked to
FR-06.1	The system shall generate a personalized study plan based on courses, deadlines, and student availability.	Yomna Mohamed Fekry	CR-01, CR-04
FR-06.2	The system shall distribute study tasks across available time slots while respecting priorities.	Yomna Mohamed Fekry	CR-04
FR-06.3	The system shall allow a student to regenerate the study plan after an availability or deadline change.	Yomna Mohamed Fekry	CR-04, CR-06

ID	Requirement	Owner	Linked to
FR-06.4	The system shall reject an AI-generated study plan that does not match the expected plan structure.	Yomna Mohamed Fekry	CR-04
FR-06.5	The system shall retain the previously saved study plan when an AI-generated plan is rejected.	Yomna Mohamed Fekry	CR-04

FR-07 — Enable Smart Scheduling

Owner: Yomna Mohamed Fekry | **Linked to:** CR-04, CR-05

ID	Requirement	Owner	Linked to
FR-07.1	The system shall schedule tasks based on deadlines, priorities, workload, availability, and dependencies.	Yomna Mohamed Fekry	CR-04
FR-07.2	The system shall detect scheduling conflicts between tasks and overlapping time slots.	Yomna Mohamed Fekry	CR-04
FR-07.3	The system shall suggest alternative time slots when scheduling conflicts occur.	Yomna Mohamed Fekry	CR-05
FR-07.4	The system shall allow a student to manually adjust scheduled tasks and times.	Yomna Mohamed Fekry	CR-04
FR-07.5	The system shall allow a student to mark a scheduled task as completed or not completed.	Yomna Mohamed Fekry	CR-04, CR-06

FR-08 — Enable Adaptive Planning

Owner: Yomna Mohamed Fekry | **Linked to:** CR-04, CR-05, CR-06, CR-11

ID	Requirement	Owner	Linked to
FR-08.1	The system shall detect when a student is falling behind on tasks or deadlines.	Yomna Mohamed Fekry	CR-06
FR-08.2	The system shall recommend rescheduling tasks when the student is behind schedule.	Yomna Mohamed Fekry	CR-04, CR-05
FR-08.3	The system shall recommend reprioritizing tasks when the student is behind schedule.	Yomna Mohamed Fekry	CR-05
FR-08.4	The system shall require student approval before applying adaptive plan changes.	Yomna Mohamed Fekry	CR-11
FR-08.5	The system shall allow a student to accept adaptive planning recommendations.	Yomna Mohamed Fekry	CR-11
FR-08.6	The system shall allow a student to reject adaptive planning recommendations.	Yomna Mohamed Fekry	CR-11

ID	Requirement	Owner	Linked to
FR-08.7	The system shall allow a student to modify adaptive planning recommendations.	Yomna Mohamed Fekry	CR-11

FR-09 — Manage Career Roadmaps

Owner: Arwa Mohey Hamdy | **Linked to:** CR-07, CR-09

ID	Requirement	Owner	Linked to
FR-09.1	The system shall allow a student to select a target career from a predefined list.	Arwa Mohey Hamdy	CR-07
FR-09.2	The system shall allow a student to enter a custom target career.	Arwa Mohey Hamdy	CR-07
FR-09.3	The system shall generate a career roadmap with required skills, learning steps, and milestones.	Arwa Mohey Hamdy	CR-07
FR-09.4	The system shall allow a student to track progress toward completing career roadmap milestones.	Arwa Mohey Hamdy	CR-09
FR-09.5	The system shall allow a student to update their target career.	Arwa Mohey Hamdy	CR-09
FR-09.6	The system shall regenerate the career roadmap after the target career is updated.	Arwa Mohey Hamdy	CR-09

FR-10 — Track Skills

Owner: Arwa Mohey Hamdy | **Linked to:** CR-08, CR-09

ID	Requirement	Owner	Linked to
FR-10.1	The system shall allow a student to add technical and soft skills with current skill level.	Arwa Mohey Hamdy	CR-08
FR-10.2	The system shall compare a student's skills against the required skills of their target career.	Arwa Mohey Hamdy	CR-09
FR-10.3	The system shall identify skill gaps based on career requirements.	Arwa Mohey Hamdy	CR-09
FR-10.4	The system shall allow a student to update skill levels as they progress.	Arwa Mohey Hamdy	CR-09

FR-11 — Manage Projects and Portfolios

Owner: Arwa Mohey Hamdy | **Linked to:** CR-08

ID	Requirement	Owner	Linked to
FR-11.1	The system shall allow a student to add a project with title, description, technologies, and completion status.	Arwa Mohey Hamdy	CR-08
FR-11.2	The system shall allow a student to link projects to the skills they demonstrate.	Arwa Mohey Hamdy	CR-08
FR-11.3	The system shall generate a portfolio view of the student's completed projects.	Arwa Mohey Hamdy	CR-08
FR-11.4	The system shall allow a student to add project links and documentation to their portfolio.	Arwa Mohey Hamdy	CR-08

FR-12 — Manage Internship and Job Preparation

Owner: Arwa Mohey Hamdy | **Linked to:** CR-07, CR-08

ID	Requirement	Owner	Linked to
FR-12.1	The system shall allow a student to track internship and job applications with company, position, status, and deadline.	Arwa Mohey Hamdy	CR-07
FR-12.2	The system shall allow a student to track interview preparation tasks for each application.	Arwa Mohey Hamdy	CR-07
FR-12.3	The system shall allow a student to track CV updates related to applications.	Arwa Mohey Hamdy	CR-08
FR-12.4	The system shall allow a student to track portfolio updates related to applications.	Arwa Mohey Hamdy	CR-08
FR-12.5	The system shall allow a student to record the outcome of an application.	Arwa Mohey Hamdy	CR-07
FR-12.6	The system shall allow a student to record feedback received for an application.	Arwa Mohey Hamdy	CR-07

FR-13 — Manage Habits

Owner: Mostafa Khamis Abozead | **Linked to:** CR-03

ID	Requirement	Owner	Linked to
FR-13.1	The system shall allow a student to create a habit with name, frequency, goal, and reminder settings.	Mostafa Khamis Abozead	CR-03
FR-13.2	The system shall allow a student to mark a habit as completed for a scheduled day.	Mostafa Khamis Abozead	CR-03

ID	Requirement	Owner	Linked to
FR-13.3	The system shall track habit streaks based on consecutive completions.	Mostafa Khamis Abozead	CR-03
FR-13.4	The system shall allow a student to link habits to academic goals.	Mostafa Khamis Abozead	CR-03

FR-14 — Analyze Study Workload

Owner: Mostafa Khamis Abozead | **Linked to:** CR-06

ID	Requirement	Owner	Linked to
FR-14.1	The system shall calculate the remaining workload of each active goal from its unfinished scheduled tasks and their scheduled durations.	Mostafa Khamis Abozead	CR-06
FR-14.2	The system shall calculate the remaining available study time of each active goal up to its target date, using the student's availability.	Mostafa Khamis Abozead	CR-06
FR-14.3	The system shall calculate the percentage of planned task duration completed for each active goal.	Mostafa Khamis Abozead	CR-06
FR-14.4	The system shall display the remaining workload, remaining available study time, and the percentage of planned task duration completed for each active goal.	Mostafa Khamis Abozead	CR-06
FR-14.5	The system shall mark an active goal as overloaded when its remaining workload exceeds its remaining available study time.	Mostafa Khamis Abozead	CR-06

FR-15 — Predict and Warn Early

Owner: Mostafa Khamis Abozead | **Linked to:** CR-06, CR-10, CR-11

ID	Requirement	Owner	Linked to
FR-15.1	The system shall estimate the likelihood of completing each active goal before its target date from its percentage of planned task duration completed and its remaining available study time.	Mostafa Khamis Abozead	CR-06
FR-15.2	The system shall generate an early warning for an active goal before its target date when the goal is marked as overloaded.	Mostafa Khamis Abozead	CR-06, CR-10

ID	Requirement	Owner	Linked to
FR-15.3	The system shall identify which scheduled tasks of an active goal push its workload over its available study time.	Mostafa Khamis Abozead	CR-06, CR-10
FR-15.4	The system shall suggest recovery actions for a goal with an active early warning so the student can recover before the target date.	Mostafa Khamis Abozead	CR-11

FR-16 — Coach Students with AI

Owner: Mostafa Khamis Abozead | **Linked to:** CR-06, CR-10, CR-11, CR-12

ID	Requirement	Owner	Linked to
FR-16.1	The system shall provide personalized recommendations based on the student's goals, scheduled tasks, habits, and workload progress.	Mostafa Khamis Abozead	CR-06
FR-16.2	The system shall offer study advice tailored to upcoming deadlines and workload.	Mostafa Khamis Abozead	CR-06, CR-10
FR-16.3	The system shall suggest recovery strategies when a student falls behind their plan.	Mostafa Khamis Abozead	CR-11
FR-16.4	The system shall provide motivational guidance to help students stay on track with their goals.	Mostafa Khamis Abozead	CR-12

FR-17 — Display Student Dashboard

Owner: Muhammed Hany Hamdy | **Linked to:** CR-01, CR-02, CR-03, CR-06, CR-07

ID	Requirement	Owner	Linked to
FR-17.1	The system shall display today's tasks and upcoming deadlines on the student dashboard.	Muhammed Hany Hamdy	CR-02
FR-17.2	The system shall display the current status of each goal on the dashboard.	Muhammed Hany Hamdy	CR-02
FR-17.3	The system shall display the completion percentage of each goal on the dashboard.	Muhammed Hany Hamdy	CR-01, CR-02
FR-17.4	The system shall display habit streaks and completion status on the dashboard.	Muhammed Hany Hamdy	CR-03

ID	Requirement	Owner	Linked to
FR-17.5	The system shall display the study workload summary of each active goal on the dashboard.	Muhammed Hany Hamdy	CR-06
FR-17.6	The system shall display career progress on the dashboard.	Muhammed Hany Hamdy	CR-07

FR-18 — Send Notifications

Owner: Muhammed Hany Hamdy | **Linked to:** CR-02, CR-03, CR-04, CR-05, CR-10

ID	Requirement	Owner	Linked to
FR-18.1	The system shall send reminders for upcoming assignments and exam deadlines.	Muhammed Hany Hamdy	CR-02
FR-18.2	The system shall send reminders for scheduled study tasks.	Muhammed Hany Hamdy	CR-04, CR-05
FR-18.3	The system shall send reminders for scheduled habits.	Muhammed Hany Hamdy	CR-03
FR-18.4	The system shall send early warning notifications when a student is at risk of missing deadlines.	Muhammed Hany Hamdy	CR-10
FR-18.5	The system shall send goal milestone reminders.	Muhammed Hany Hamdy	CR-02
FR-18.6	The system shall notify a student when a habit streak reaches a new milestone.	Muhammed Hany Hamdy	CR-03

FR-19 — Manage Study Groups and Accountability

Owner: Muhammed Hany Hamdy | **Linked to:** CR-12

ID	Requirement	Owner	Linked to
FR-19.1	The system shall allow a student to create a study group.	Muhammed Hany Hamdy	CR-12
FR-19.2	The system shall allow a student to join a study group.	Muhammed Hany Hamdy	CR-12
FR-19.3	The system shall allow study group members to share goals.	Muhammed Hany Hamdy	CR-12
FR-19.4	The system shall allow study group members to track group progress.	Muhammed Hany Hamdy	CR-12
FR-19.5	The system shall allow a student to find an accountability partner.	Muhammed Hany Hamdy	CR-12

ID	Requirement	Owner	Linked to
FR-19.6	The system shall allow a student to connect with an accountability partner.	Muhammed Hany Hamdy	CR-12
FR-19.7	The system shall allow study groups to share progress updates.	Muhammed Hany Hamdy	CR-12
FR-19.8	The system shall allow study groups to share support messages.	Muhammed Hany Hamdy	CR-12

FR-20 — Manage Administration

Owner: Muhammed Hany Hamdy | **Linked to:** CR-13, CR-14, CR-15, CR-16

ID	Requirement	Owner	Linked to
FR-20.1	The system shall allow an administrator to view student account details.	Muhammed Hany Hamdy	CR-13
FR-20.2	The system shall allow an administrator to create a student account.	Muhammed Hany Hamdy	CR-13
FR-20.3	The system shall allow an administrator to update student account details.	Muhammed Hany Hamdy	CR-13
FR-20.4	The system shall allow an administrator to suspend a student account.	Muhammed Hany Hamdy	CR-13
FR-20.5	The system shall provide usage statistics showing the number of registered students.	Muhammed Hany Hamdy	CR-14
FR-20.6	The system shall provide usage statistics showing the number of active students.	Muhammed Hany Hamdy	CR-14
FR-20.7	The system shall provide usage statistics showing platform feature usage.	Muhammed Hany Hamdy	CR-14
FR-20.8	The system shall allow an administrator to investigate student account issues.	Muhammed Hany Hamdy	CR-15
FR-20.9	The system shall allow an administrator to resolve student account issues.	Muhammed Hany Hamdy	CR-15
FR-20.10	The system shall display platform retention metrics showing student activity over time.	Muhammed Hany Hamdy	CR-16

Traceability Notes

- **FR-01**: Profile management (including availability, used by scheduling) is mapped to CR-01, the only documented need covering planning courses and workload.

- **FR-02**: Goal lifecycle is mapped to CR-01 (planning) and CR-02 (knowing what is due and when): creating a goal with a target date and priority maps to both, the target date and the completed state map to CR-02, and updating the details, archiving, and deleting map to CR-01. B3 has no CR dedicated to goal tracking, so CR-01 and CR-02 are the closest documented needs for these rows.
- **FR-03.2**: Narrowed to course credits and weekly workload, which are covered by CR-01 and CR-04. Grade recording was removed because no CR in B3 supports grades.
- **FR-03.3**: Course progress is mapped to CR-02 (outstanding assignments) and CR-04 (workload in one realistic schedule).
- **FR-04.3 / FR-04.4**: Assignment status tracking is mapped to CR-02 and CR-04; exam preparation reminders are mapped to CR-02, since CR-05 concerns prioritisation when deadlines collide.
- **FR-05**: Review, edit milestones, and edit tasks were split into FR-05.3, FR-05.4, and FR-05.5; the approval behaviour is FR-05.6. Review and edit are mapped to CR-04 because the student must be able to fit the AI proposal into a realistic schedule.
- **FR-06.1 / FR-06.3**: Plan generation is mapped to CR-01 and CR-04; regeneration after an availability or deadline change is also mapped to CR-06, since a changed workload may no longer be finishable.
- **FR-06.4 / FR-06.5**: Split into rejecting an invalid AI-generated plan and retaining the previously saved plan. Both are mapped to CR-04, because the student must keep a plan that still fits courses, projects, and work into one realistic schedule when a generation attempt fails.
- **FR-07**: Conflict resolution alternatives are mapped to CR-05 (knowing what to do first when deadlines collide); manual adjustment is mapped to CR-04 (fitting the work into one realistic schedule).
- **FR-08.2 / FR-08.3**: Rescheduling is mapped to CR-04 and CR-05; reprioritising is mapped to CR-05. Accept, reject, and modify are mapped to CR-11, the documented need for a clear way to recover after a plan breaks.
- **FR-09 / FR-10**: Career roadmap and skill work are mapped to CR-07 (balancing coursework with job preparation) and CR-09 (knowing which skills are missing for the target career). FR-10.1 is mapped to CR-08, since recorded skills are the content the student shows to employers in one place.
- **FR-11 / FR-12**: Projects, portfolio, CV, and portfolio updates are mapped to CR-08 (showing skills and projects to employers in one place); application tracking, interview preparation, outcomes, and feedback are mapped to CR-07 (job preparation).
- **FR-13.4**: Linking habits to academic goals is mapped to CR-03, the documented need to build good study habits from the start, so FR-13 is now linked to CR-03 only.
- **FR-14 (narrowed)**: Previously this FR covered grade trends, assignment completion rates, and GPA progress, which no CR in B3 supports. It is now **Analyze Study Workload**, defined only by the workload-planning need in CR-06 (knowing early when the workload is too much to finish): remaining workload, remaining available study time, the percentage of planned task duration completed, and the overloaded state.
- **FR-15**: Reworked around the same workload need. Early warnings are raised for goals already marked as overloaded, which is exactly the early warning CR-10 describes; recovery suggestions are mapped to CR-11.
- **FR-16.1 / FR-16.2**: Coaching recommendations and deadline-aware study advice are mapped to CR-06 and CR-10; recovery strategies to CR-11; motivational guidance to CR-12.
- **FR-17**: Renumbered after the splits. Today's tasks and upcoming deadlines are mapped to CR-02 (knowing what is due and when). FR-17.2 shows the current status of each goal, mapped to CR-02, because the status is read against the goal's target date. FR-17.3 shows the completion percentage of each goal, mapped to CR-01 (progress against the planned work) and CR-02 (progress against the goal's target date). As with FR-02, B3 has no CR dedicated to goal tracking, so CR-01 and CR-02 are the

closest documented needs for these two rows. Habit streaks are mapped to CR-03 and career progress to CR-07. FR-17.5 previously showed an academic performance summary, which had no supporting CR; it now shows the study workload summary under CR-06.

- **FR-18**: Reminders for scheduled study tasks are mapped to CR-04 and CR-05; habit reminders and habit streak milestones to CR-03; deadline reminders to CR-02; early warning notifications to CR-10.
- **FR-19**: Creating and joining a study group, and finding and connecting with an accountability partner, were split into separate behaviours, all mapped to CR-12.
- **FR-20**: Account management is mapped to CR-13; issue investigation and resolution to CR-15; registered students, active students, and platform feature usage statistics to CR-14; retention metrics to CR-16.

Scope notes

- Grade trends, assignment completion rates, GPA progress, and grade recording are **not** part of B4, because no Customer Requirement in B3 supports them. If the team wants these capabilities, a customer requirement must be added to B3 first, and FR-14 plus FR-17.5 can then be extended accordingly.
- Career, job-preparation, and study-group requirements are kept because the assignment requires four high-level FRs per member and the README assigns these areas to Members 3 and 5. They remain later-phase items against the suggested MVP and must be prioritised in the backlog (C8).

B5 — Non-Functional Requirements

ID	Category	Requirement	Conditions	How measured
NFR-01	Performance	The API shall respond within **500 ms for at least 95% of ordinary non-AI API requests**. [Documented — README § Quality Attributes]	Release-candidate build in the team's test environment with a fixed dataset and authenticated sessions, under 50 concurrent virtual users for 10 minutes, excluding AI-provider calls.	**Method:** Scripted load test recording server-side response time per request as the 95th percentile; **Pass** when the 95th percentile is at or below 500 ms, **Fail** otherwise.
NFR-02	Security	Every protected student-data endpoint shall require authentication and enforce authorization, with **zero unauthorized data access** to any student record, and administrator-only endpoints shall reject student accounts. [Documented — README § Quality Attributes]	Test environment seeded with two students and one administrator. Verify that missing, invalid, and non-owner student tokens are rejected, student accounts are denied access to administrator-only endpoints, and valid administrator access to administrator-only endpoints succeeds.	**Method:** Automated API-level integration tests, one per case, recording the actual result and whether another student's record was returned or changed; **Pass** when every case returns its expected result with no unauthorized access, **Fail** otherwise.

ID	Category	Requirement	Conditions	How measured
NFR-03	Reliability	When an AI-provider request fails or returns schema-invalid output, the system shall **not lose, overwrite, or corrupt saved student data and shall not save a partial plan**, returning a defined error instead. [Documented — README § Quality Attributes]	Failure scenarios (provider error, rate limit, unavailable service, schema-invalid output) applied to both AI write paths, with the student's saved goal, study plan, and scheduled tasks snapshotted beforehand.	**Method:** Automated integration tests with the AI provider replaced by a test double, comparing the student's saved records before and after each failed request; **Pass** when every test returns a defined error and the saved records are unchanged, **Fail** otherwise.
NFR-04	Usability	In a usability test, first-time students shall create a goal in natural language and review the AI-generated plan (milestones, tasks, and deadlines). Success targets: at least 4 of 5 participants finish within 5 minutes, with no more than 1 moderator intervention. [Task documented — README § Quality Attributes]	Moderated test of the running web frontend with 5 students who have not used Aspirra before, given only the task statement: enter a goal, then review the proposed plan, with the AI response stubbed and any moderator guidance beyond restating the task counted as an intervention.	**Method:** One timed, recorded session per participant, logging elapsed time, completion, and interventions; **Pass** when at least 4 of 5 participants finish within 5 minutes with no more than 1 intervention, **Fail** otherwise.

B6 — Subsystems, Interfaces & Traceability

System decomposition (system → containers → components)

Level	Element	Description
System	Aspirra	AI-powered student planning and career platform
Container	Web Frontend	Student and administrator web application (React / Next.js)
Container	Backend API	Application logic (ASP.NET Core); its components are the 10 subsystems below
Container	Relational Database	Stores the data of every subsystem
External system	External: AI Provider	LLM API used for goal planning, study plans, career roadmaps and coaching

Level	Element	Description
External system	External: Email Service	Delivers reminders and warnings

Subsystems with their FRs

Each subsystem groups related high-level FRs. Each FR is one service, with its own feature branch in the repository and its own Jira label.

ID	Subsystem	High-level FRs	Services
SUB-01	Student Profile	FR-01 Manage Student Profile	student-service
SUB-02	Academic Management	FR-02 Manage Academic Goals FR-03 Manage Courses FR-04 Manage Assignments and Exams	academic-service course-service assessment-service
SUB-03	AI Planning & Scheduling	FR-05 Plan Goals with AI FR-06 Generate Study Plans FR-07 Enable Smart Scheduling FR-08 Enable Adaptive Planning	goal-planner-service study-planner-service scheduling-service adaptation-service
SUB-04	Career Development	FR-09 Manage Career Roadmaps FR-10 Track Skills FR-11 Manage Projects and Portfolios FR-12 Manage Internship and Job Preparation	career-service skill-service portfolio-service job-preparation-service
SUB-05	Habit Tracking	FR-13 Manage Habits	habit-service
SUB-06	Insights & Coaching	FR-14 Analyze Study Workload FR-15 Predict and Warn Early FR-16 Coach Students with AI	analytics-service prediction-service ai-coach-service
SUB-07	Student Dashboard	FR-17 Display Student Dashboard	dashboard-service
SUB-08	Notifications	FR-18 Send Notifications	notification-service
SUB-09	Study Groups & Accountability	FR-19 Manage Study Groups and Accountability	study-group-service
SUB-10	Administration	FR-20 Manage Administration	administration-service

Interfaces

Interface	Provided by	What it offers
IStudentProfile	SUB-01 Student Profile	Create, update and view the profile (FR-01); the student's availability and interests for other subsystems
IAcademic	SUB-02 Academic Management	Goals (FR-02), courses (FR-03), assignments, exams and their deadlines (FR-04)
IPlanning	SUB-03 AI Planning & Scheduling	AI goal plans (FR-05), study plans (FR-06), scheduled tasks and conflicts (FR-07), adaptive recommendations (FR-08)
ICareer	SUB-04 Career Development	Career roadmap and its progress (FR-09), skills and skill gaps (FR-10), projects and portfolio (FR-11), job applications (FR-12)
IHabit	SUB-05 Habit Tracking	Habits, completions and streaks (FR-13)
IInsights	SUB-06 Insights & Coaching	Workload per goal and overload state (FR-14), completion likelihood and early warnings (FR-15), coaching advice (FR-16)
IDashboard	SUB-07 Student Dashboard	The student dashboard view (FR-17)
INotification	SUB-08 Notifications	Reminders and notifications (FR-18)
IStudyGroup	SUB-09 Study Groups & Accountability	Groups, shared goals and progress, partners and messages (FR-19)
IAdministration	SUB-10 Administration	Account management, statistics and retention metrics (FR-20)
IAIProvider	External: AI Provider	Generates goal plans, study plans, career roadmaps and coaching text
IEmailGateway	External: Email Service	Sends emails

Traceability matrix

Subsystem	High-level FRs	Provided interface	Required interfaces
SUB-01 Student Profile	FR-01 Manage Student Profile	IStudentProfile	—
SUB-02 Academic Management	FR-02 Manage Academic Goals, FR-03 Manage Courses, FR-04 Manage Assignments and Exams	IAcademic	IStudentProfile
SUB-03 AI Planning & Scheduling	FR-05 Plan Goals with AI, FR-06 Generate Study Plans, FR-07 Enable Smart Scheduling, FR-08 Enable Adaptive Planning	IPlanning	IStudentProfile, IAcademic, IAIProvider (External)

Subsystem	High-level FRs	Provided interface	Required interfaces
SUB-04 Career Development	FR-09 Manage Career Roadmaps, FR-10 Track Skills, FR-11 Manage Projects and Portfolios, FR-12 Manage Internship and Job Preparation	ICareer	IStudentProfile, IAIProvider (External)
SUB-05 Habit Tracking	FR-13 Manage Habits	IHabit	IAcademic
SUB-06 Insights & Coaching	FR-14 Analyze Study Workload, FR-15 Predict and Warn Early, FR-16 Coach Students with AI	IInsights	IStudentProfile, IAcademic, IPlanning, IHabit, IAIProvider (External)
SUB-07 Student Dashboard	FR-17 Display Student Dashboard	IDashboard	IAcademic, IPlanning, ICareer, IHabit, IInsights
SUB-08 Notifications	FR-18 Send Notifications	INotification	IAcademic, IPlanning, IHabit, IInsights, IEmailGateway (External)
SUB-09 Study Groups & Accountability	FR-19 Manage Study Groups and Accountability	IStudyGroup	IStudentProfile, IAcademic, IInsights
SUB-10 Administration	FR-20 Manage Administration	IAdministration	IStudentProfile

B7 — Use Case Diagram

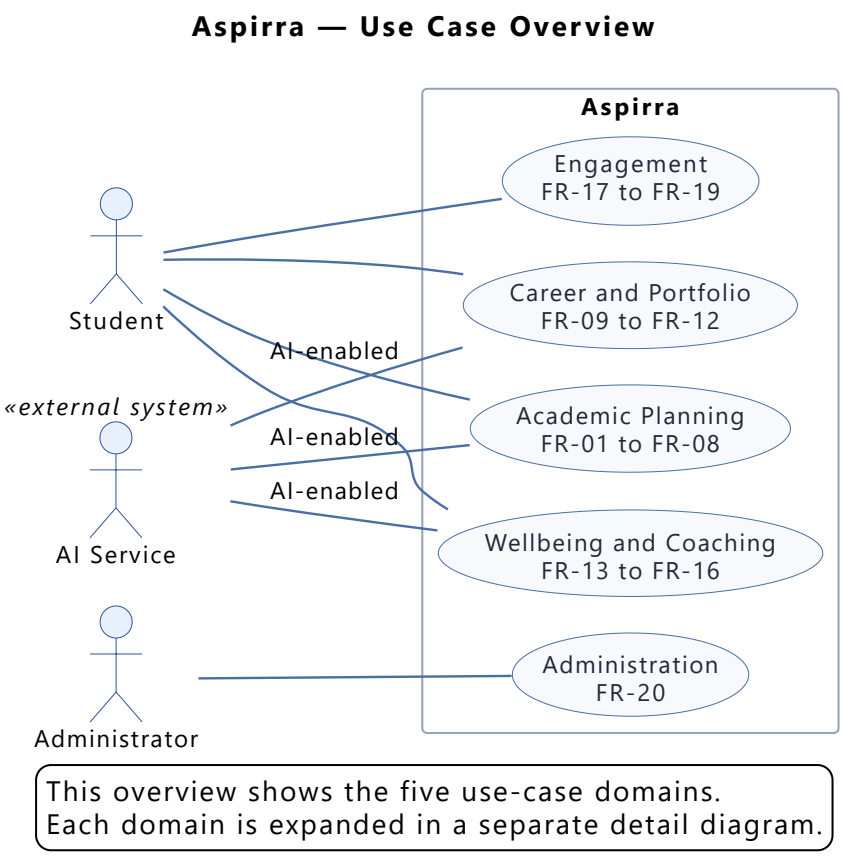
Diagram views

The use case model is split into an overview and five detail diagrams so that each part stays readable.

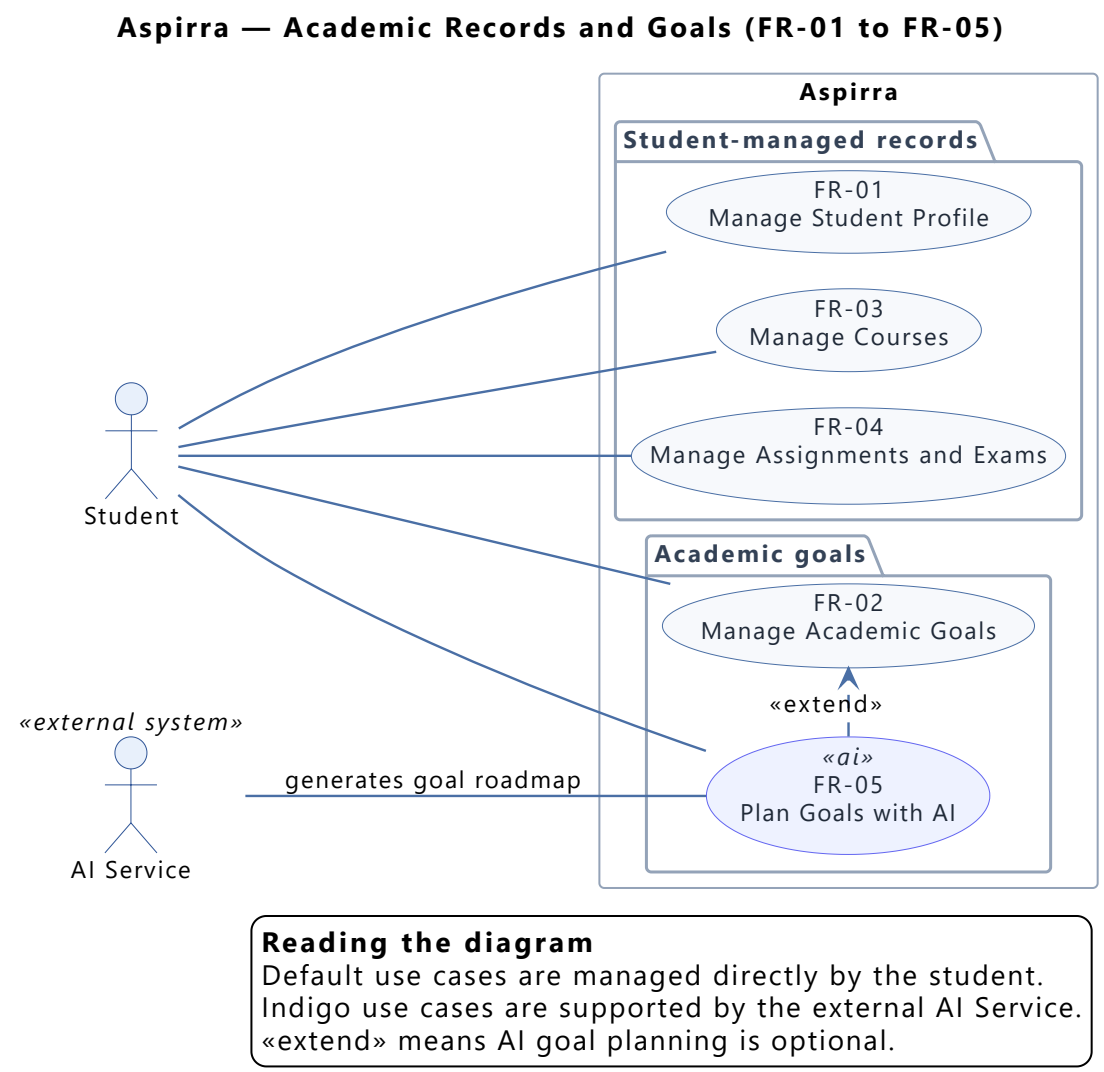
View	Scope	PlantUML source
Overview	Actors and the five use-case domains	`b7-use-case-overview.puml`
Academic records and goals	FR-01 to FR-05	`b7-use-case-academic-records.puml`
Study planning and adaptation	FR-06 to FR-08, with shared FR-14 and FR-15	`b7-use-case-study-planning.puml`
Career and portfolio	FR-09 to FR-12	`b7-use-case-career-portfolio.puml`
Wellbeing and coaching	FR-13 to FR-16	`b7-use-case-wellbeing-coaching.puml`

View	Scope	PlantUML source
Engagement and administration	FR-17 to FR-20, with shared FR-14	`b7-use-case-engagement-administration.puml`

Overview



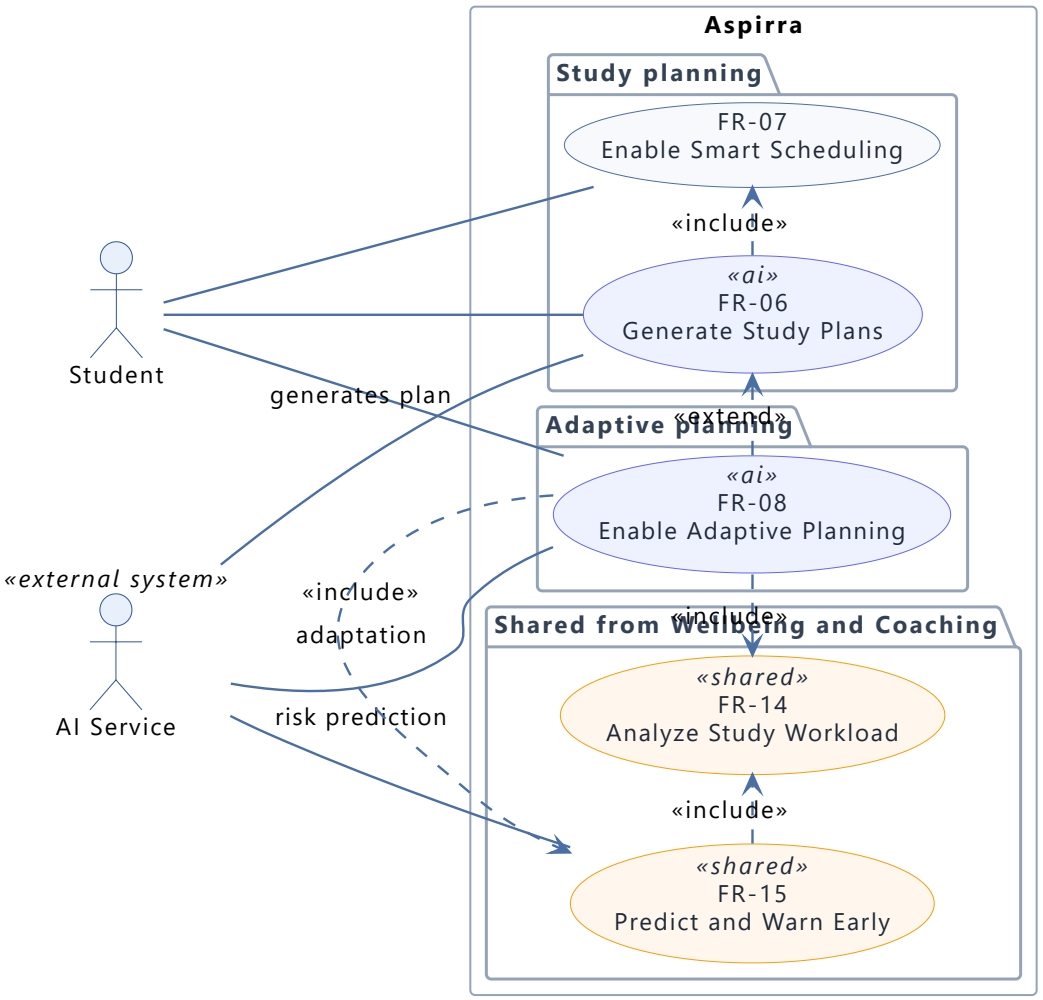
Academic records and goals



AI goal planning is optional and extends manual academic-goal management.

Study planning and adaptation

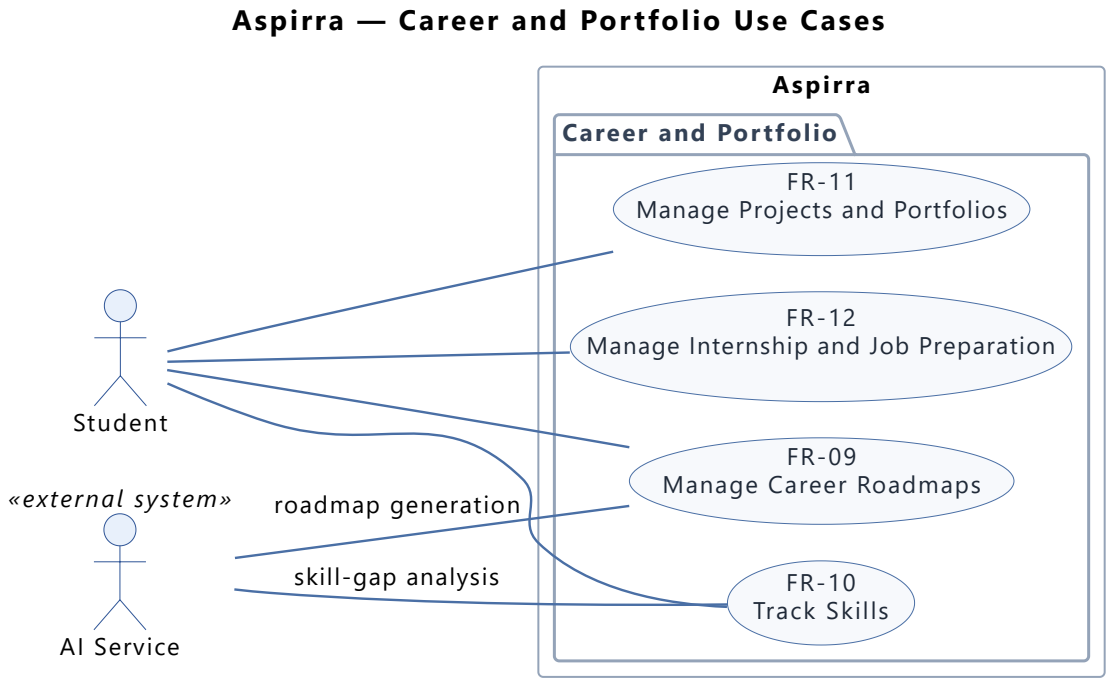
Aspirra — Study Planning and Adaptation (FR-06 to FR-08)



Reading the diagram
Indigo use cases use the external AI Service.
Amber use cases are shared with the Wellbeing and Coaching view.
«include» = required behaviour; «extend» = optional behaviour.

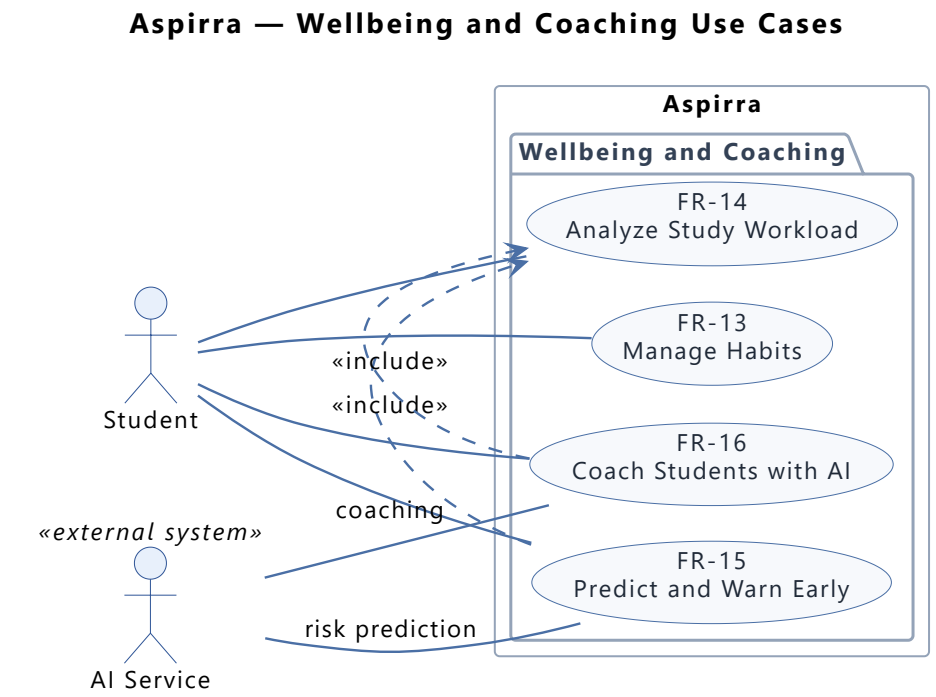
FR-14 and FR-15 are repeated here as shared use cases to show the adaptive-planning dependencies.

Career and portfolio



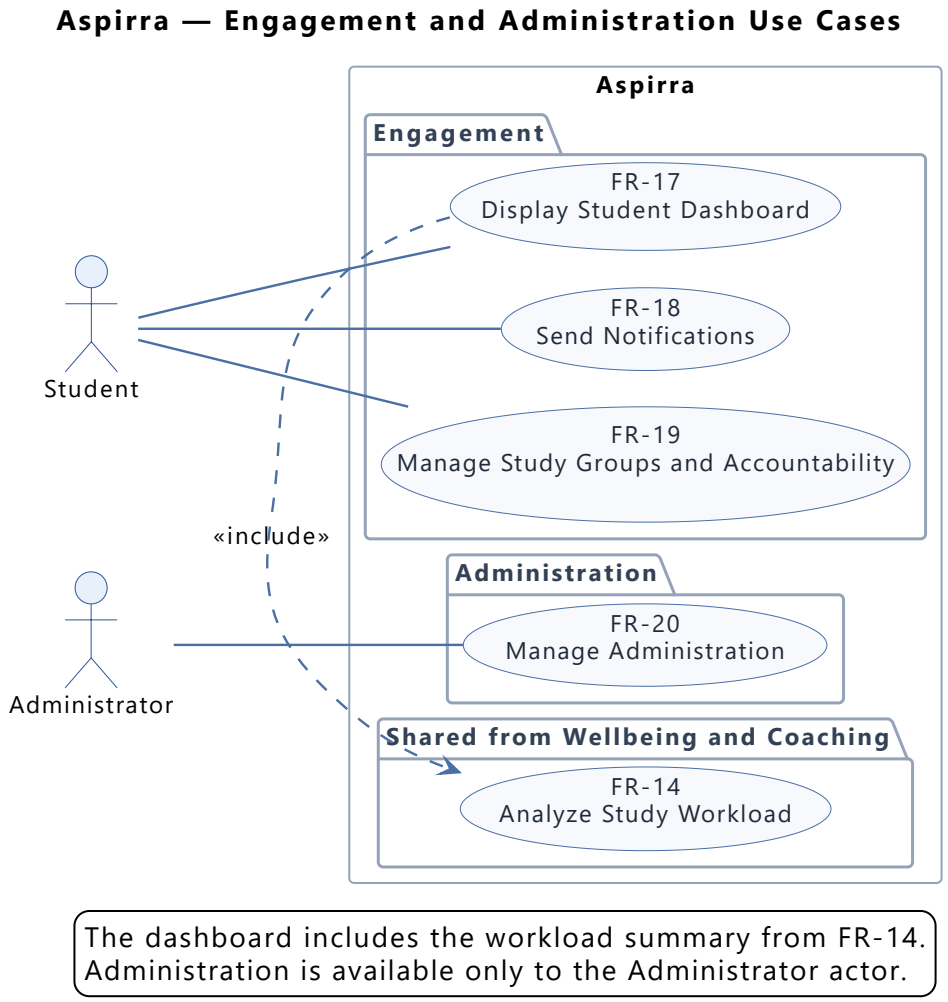
The AI Service supports career-roadmap generation and skill-gap analysis.

Wellbeing and coaching



Prediction and coaching both use the workload analysis in FR-14.

Engagement and administration



FR-14 is repeated here because the dashboard includes the workload summary.

Actors

Actor	Type	Responsibility
Student	Primary	Manages academic work, plans, habits, career preparation, and collaboration features.
Administrator	Primary	Manages student accounts, platform statistics, account issues, and retention metrics.
AI Service	Secondary (external)	Generates goal roadmaps, study and adaptive plans, career roadmaps, skill-gap insights, predictions, and coaching recommendations.
Email Service	Secondary (external)	Delivers reminders, notifications, and early warnings.

Use cases

ID	Use case	Primary actor	Supporting actor	Relationships
FR-01	Manage Student Profile	Student	—	—
FR-02	Manage Academic Goals	Student	—	Extended by FR-05
FR-03	Manage Courses	Student	—	—
FR-04	Manage Assignments and Exams	Student	—	—
FR-05	Plan Goals with AI	Student	AI Service	Extends FR-02
FR-06	Generate Study Plans	Student	AI Service	Includes FR-07; extended by FR-08
FR-07	Enable Smart Scheduling	Student	—	Included by FR-06
FR-08	Enable Adaptive Planning	Student	AI Service	Extends FR-06; includes FR-14 and FR-15
FR-09	Manage Career Roadmaps	Student	AI Service	—
FR-10	Track Skills	Student	AI Service	—
FR-11	Manage Projects and Portfolios	Student	—	—
FR-12	Manage Internship and Job Preparation	Student	—	—
FR-13	Manage Habits	Student	—	—
FR-14	Analyze Study Workload	Student	—	Included by FR-08, FR-15, FR-16, and FR-17
FR-15	Predict and Warn Early	Student	AI Service	Includes FR-14; included by FR-08
FR-16	Coach Students with AI	Student	AI Service	Includes FR-14
FR-17	Display Student Dashboard	Student	—	Includes FR-14

ID	Use case	Primary actor	Supporting actor	Relationships
FR-18	Send Notifications	Student	Email Service	—
FR-19	Manage Study Groups and Accountability	Student	—	—
FR-20	Manage Administration	Administrator	—	—

Relationships

- **Extend:** AI goal planning extends manual academic-goal management, and adaptive planning extends the generated study plan.
- **Include:** Generating a study plan always schedules tasks; adaptive planning uses workload analysis and early warnings; prediction, coaching, and the dashboard all rely on workload analysis.

Part C — Planning & Management

Product backlog, estimates, sprint plan, PERT analysis, Jira board, and GitHub repository.

C8 — Product Backlog

Each high-level FR is one feature (one epic in Jira), broken into user stories numbered under it as in the lab notation (FR-02 → US-02.1, US-02.2, …). **Covers** lists the low-level FRs each story implements; a low-level FR that is a system rule (e.g. FR-01.4 "validate required fields") is written as an acceptance criterion of its story. **Priority** (High = MVP, then Medium, then Low) is the order used in the sprint plan (C9).

FR-01 — Manage Student Profile

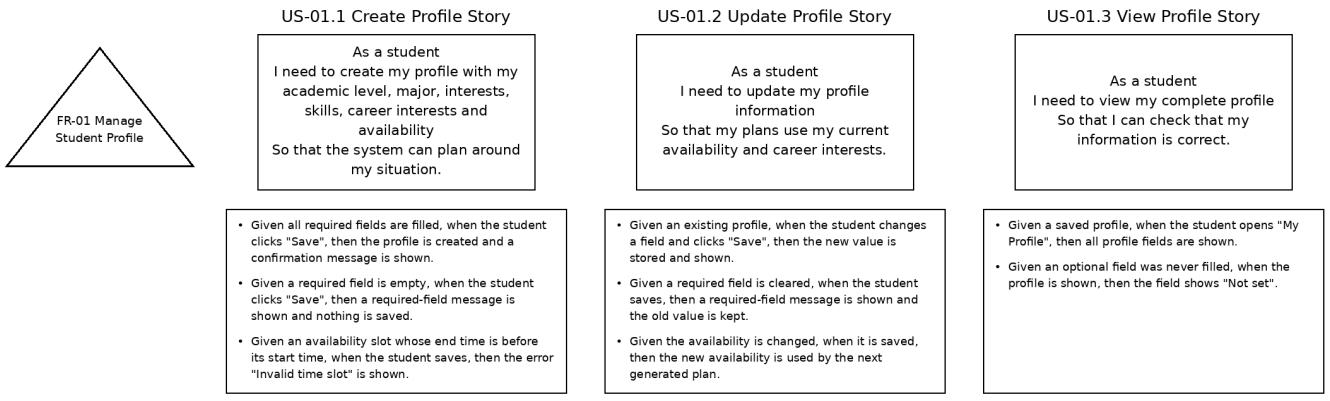


Figure 1 — Manage Student Profile feature (FR-01): user stories and their acceptance criteria.

Story	User story	Acceptance criteria	Covers	Priority
US-01.1 Create	As a student, I need to create my profile with my academic level, major, interests, skills, career	• Given all required fields are filled, when the student clicks "Save", then the profile is	FR-01.1, FR-01.4	High

Story	User story	Acceptance criteria	Covers	Priority
Profile Story	interests and availability, so that the system can plan around my situation.	created and a confirmation message is shown. • Given a required field is empty, when the student clicks "Save", then a required-field message is shown and nothing is saved. • Given an availability slot whose end time is before its start time, when the student saves, then the error "Invalid time slot" is shown.		
US-01.2 Update Profile Story	As a student, I need to update my profile information, so that my plans use my current availability and career interests.	• Given an existing profile, when the student changes a field and clicks "Save", then the new value is stored and shown. • Given a required field is cleared, when the student saves, then a required-field message is shown and the old value is kept. • Given the availability is changed, when it is saved, then the new availability is used by the next generated plan.	FR-01.2, FR-01.4	High
US-01.3 View Profile Story	As a student, I need to view my complete profile, so that I can check that my information is correct.	• Given a saved profile, when the student opens "My Profile", then all profile fields are shown. • Given an optional field was never filled, when the profile is shown, then the field shows "Not set".	FR-01.3	High

FR-02 — Manage Academic Goals

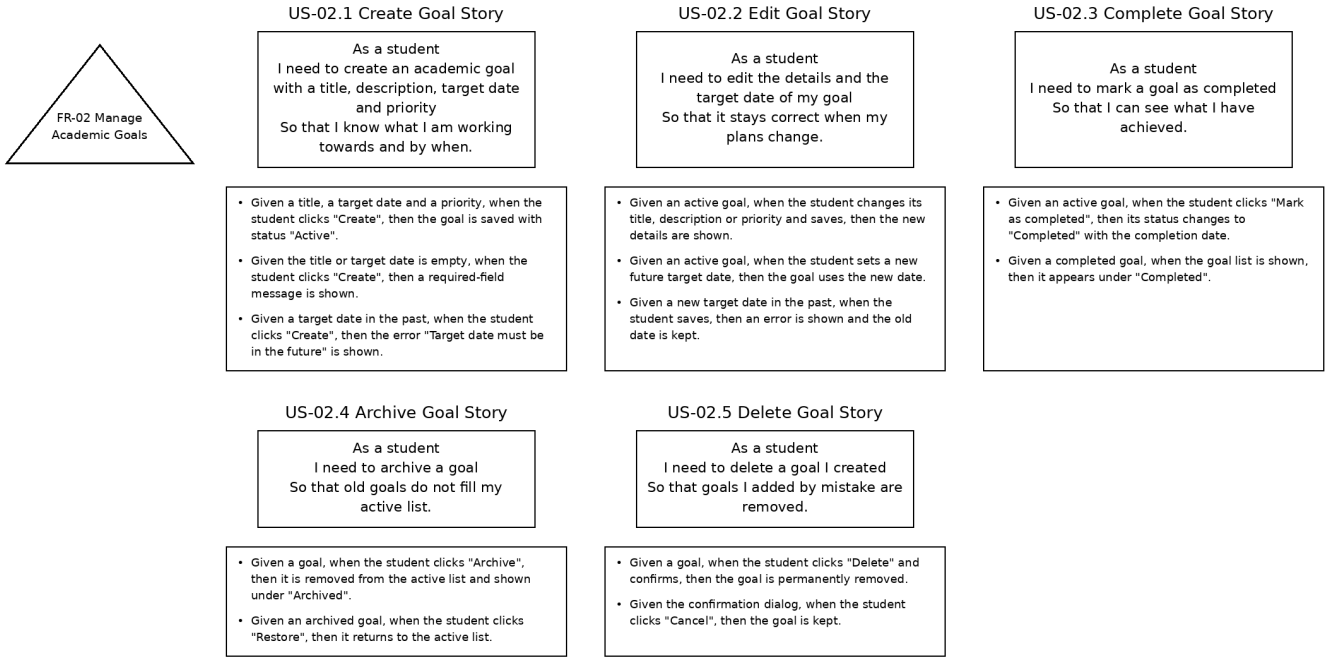


Figure 2 — Manage Academic Goals feature (FR-02): user stories and their acceptance criteria.

Story	User story	Acceptance criteria	Covers	Priority
US-02.1 Create Goal Story	As a student, I need to create an academic goal with a title, description, target date and priority, so that I know what I am working towards and by when.	• Given a title, a target date and a priority, when the student clicks "Create", then the goal is saved with status "Active". • Given the title or target date is empty, when the student clicks "Create", then a required-field message is shown. • Given a target date in the past, when the student clicks "Create", then the error "Target date must be in the future" is shown.	FR-02.1	High

Story	User story	Acceptance criteria	Covers	Priority
US-02.2 Edit Goal Story	As a student, I need to edit the details and the target date of my goal, so that it stays correct when my plans change.	• Given an active goal, when the student changes its title, description or priority and saves, then the new details are shown. • Given an active goal, when the student sets a new future target date, then the goal uses the new date. • Given a new target date in the past, when the student saves, then an error is shown and the old date is kept.	FR-02.2, FR-02.3	High
US-02.3 Complete Goal Story	As a student, I need to mark a goal as completed, so that I can see what I have achieved.	• Given an active goal, when the student clicks "Mark as completed", then its status changes to "Completed" with the completion date. • Given a completed goal, when the goal list is shown, then it appears under "Completed".	FR-02.4	High
US-02.4 Archive Goal Story	As a student, I need to archive a goal, so that old goals do not fill my active list.	• Given a goal, when the student clicks "Archive", then it is removed from the active list and shown under "Archived". • Given an archived goal, when the student clicks "Restore", then it returns to the active list.	FR-02.5	Medium
US-02.5 Delete Goal Story	As a student, I need to delete a goal I created, so that goals I added by mistake are removed.	• Given a goal, when the student clicks "Delete" and confirms, then the goal is permanently removed. • Given the confirmation dialog, when the student clicks "Cancel", then the goal is kept.	FR-02.6	Medium

FR-03 — Manage Courses

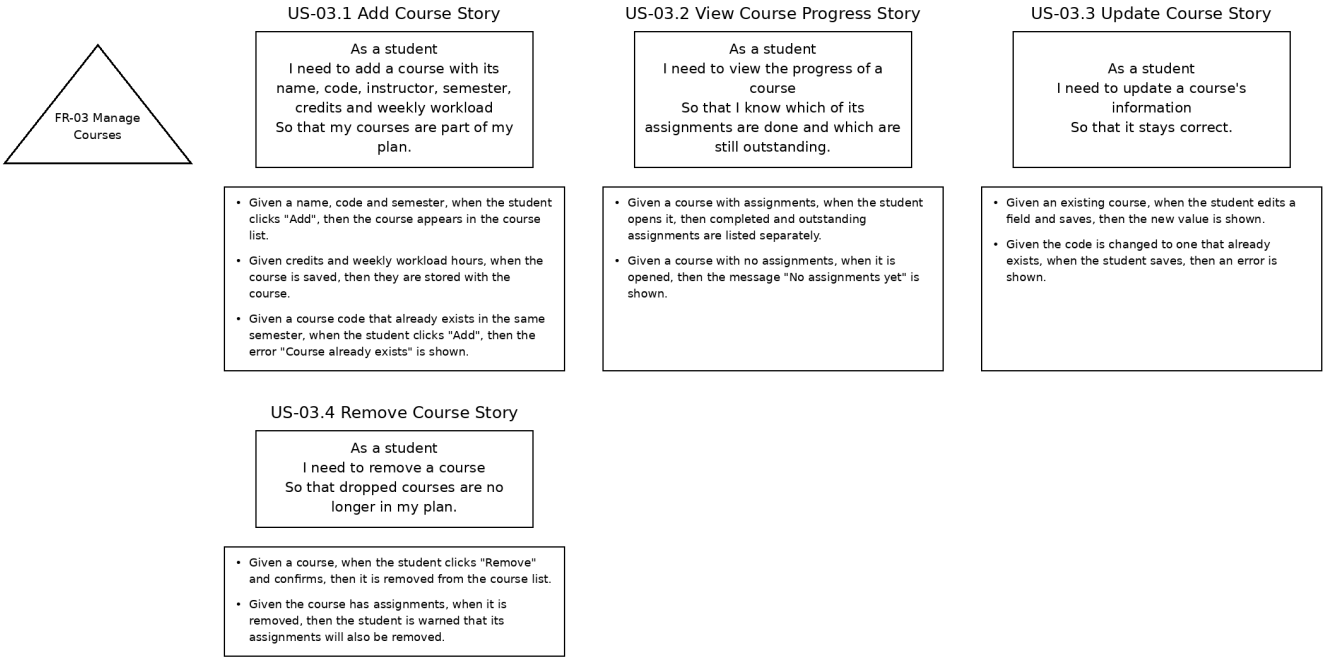


Figure 3 — Manage Courses feature (FR-03): user stories and their acceptance criteria.

Story	User story	Acceptance criteria	Covers	Priority
US-03.1 Add Course Story	As a student, I need to add a course with its name, code, instructor, semester, credits and weekly workload, so that my courses are part of my plan.	• Given a name, code and semester, when the student clicks "Add", then the course appears in the course list. • Given credits and weekly workload hours, when the course is saved, then they are stored with the course. • Given a course code that already exists in the same semester, when the student clicks "Add", then the error "Course already exists" is shown.	FR-03.1, FR-03.2	High
US-03.2 View Course Progress Story	As a student, I need to view the progress of a course, so that I know which of its assignments are done and which are still outstanding.	• Given a course with assignments, when the student opens it, then completed and outstanding assignments are listed separately. • Given a course with no assignments, when it is opened, then the message "No assignments yet" is shown.	FR-03.3	High
US-03.3 Update	As a student, I need to update a course's information, so that it stays correct.	• Given an existing course, when the student edits a field and saves, then the new value is	FR-03.4	Medium

Story	User story	Acceptance criteria	Covers	Priority
Course Story		shown. • Given the code is changed to one that already exists, when the student saves, then an error is shown.		
US-03.4 Remove Course Story	As a student, I need to remove a course, so that dropped courses are no longer in my plan.	• Given a course, when the student clicks "Remove" and confirms, then it is removed from the course list. • Given the course has assignments, when it is removed, then the student is warned that its assignments will also be removed.	FR-03.5	Medium

FR-04 — Manage Assignments and Exams

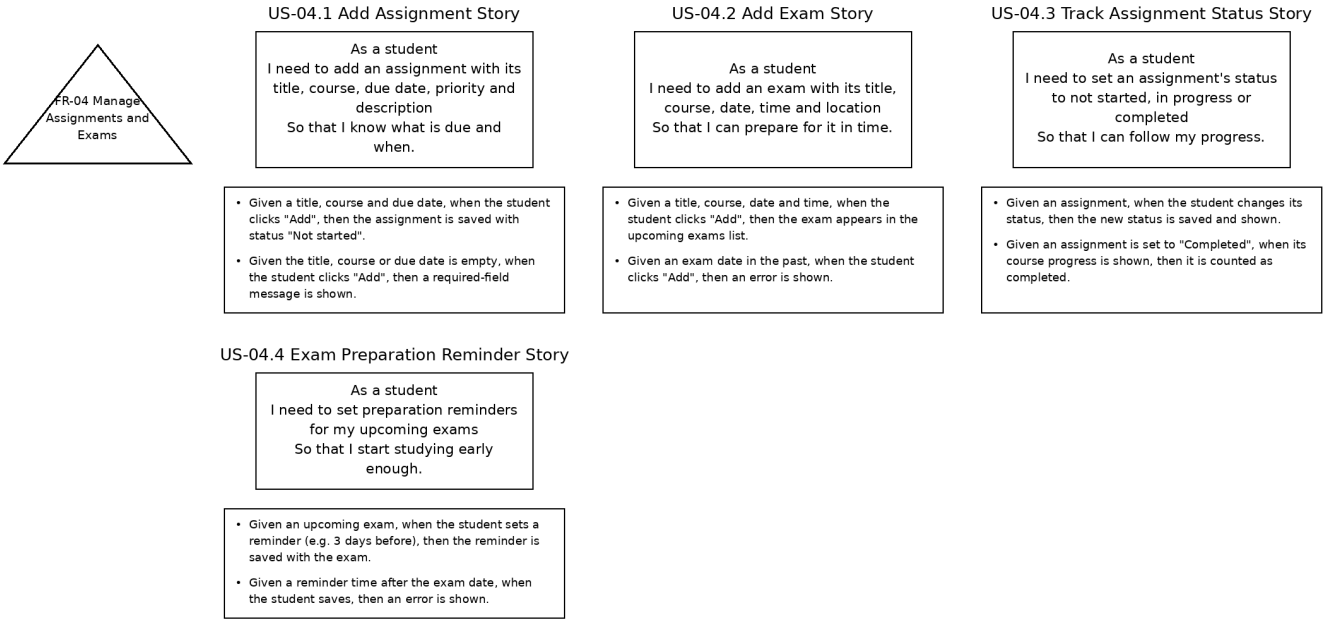


Figure 4 — Manage Assignments and Exams feature (FR-04): user stories and their acceptance criteria.

Story	User story	Acceptance criteria	Covers	Priority
US-04.1 Add Assignment Story	As a student, I need to add an assignment with its title, course, due date, priority and description, so that I know what is due and when.	• Given a title, course and due date, when the student clicks "Add", then the assignment is saved with status "Not started". • Given the title, course or due date is empty, when the student clicks "Add", then a	FR-04.1	High

Story	User story	Acceptance criteria	Covers	Priority
		required-field message is shown.		
US-04.2 Add Exam Story	As a student, I need to add an exam with its title, course, date, time and location, so that I can prepare for it in time.	• Given a title, course, date and time, when the student clicks "Add", then the exam appears in the upcoming exams list. • Given an exam date in the past, when the student clicks "Add", then an error is shown.	FR-04.2	High
US-04.3 Track Assignment Status Story	As a student, I need to set an assignment's status to not started, in progress or completed, so that I can follow my progress.	• Given an assignment, when the student changes its status, then the new status is saved and shown. • Given an assignment is set to "Completed", when its course progress is shown, then it is counted as completed.	FR-04.3	High
US-04.4 Exam Preparation Reminder Story	As a student, I need to set preparation reminders for my upcoming exams, so that I start studying early enough.	• Given an upcoming exam, when the student sets a reminder (e.g. 3 days before), then the reminder is saved with the exam. • Given a reminder time after the exam date, when the student saves, then an error is shown.	FR-04.4	Medium

FR-05 — Plan Goals with AI

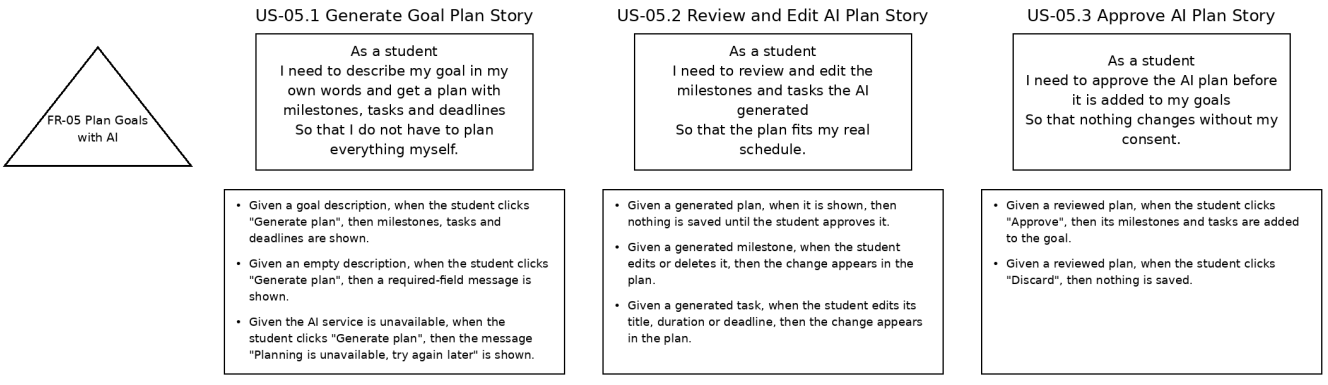


Figure 5 — Plan Goals with AI feature (FR-05): user stories and their acceptance criteria.

Story	User story	Acceptance criteria	Covers	Priority
US-05.1 Generate Goal Plan Story	As a student, I need to describe my goal in my own words and get a plan with milestones, tasks and deadlines, so that I do not have to plan everything myself.	• Given a goal description, when the student clicks "Generate plan", then milestones, tasks and deadlines are shown. • Given an empty description, when the student clicks "Generate plan", then a required-field message is shown. • Given the AI service is unavailable, when the student clicks "Generate plan", then the message "Planning is unavailable, try again later" is shown.	FR-05.1, FR-05.2	High
US-05.2 Review and Edit AI Plan Story	As a student, I need to review and edit the milestones and tasks the AI generated, so that the plan fits my real schedule.	• Given a generated plan, when it is shown, then nothing is saved until the student approves it. • Given a generated milestone, when the student edits or deletes it, then the change appears in the plan. • Given a generated task, when the student edits its title, duration or deadline, then the change appears in the plan.	FR-05.3, FR-05.4, FR-05.5	High
US-05.3 Approve AI Plan Story	As a student, I need to approve the AI plan before it is added to my goals, so that nothing changes without my consent.	• Given a reviewed plan, when the student clicks "Approve", then its milestones and tasks are added to the goal. • Given a reviewed plan, when the student clicks "Discard", then nothing is saved.	FR-05.6	High

FR-06 — Generate Study Plans

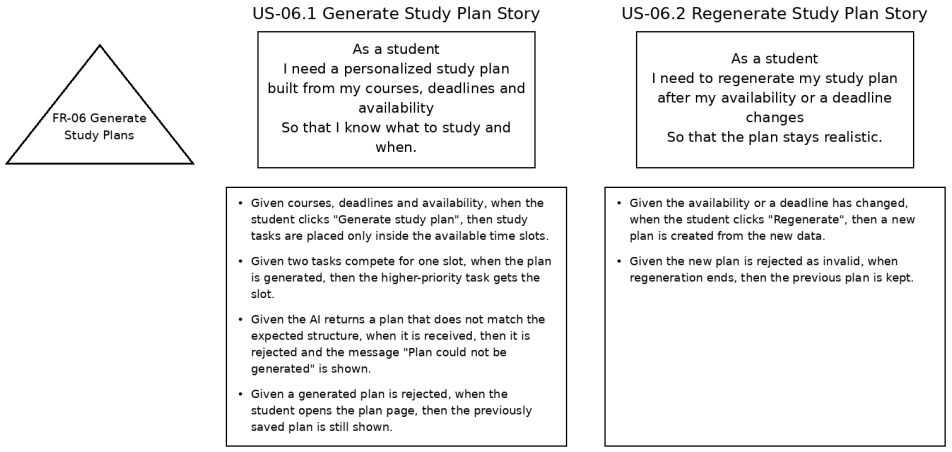


Figure 6 — Generate Study Plans feature (FR-06): user stories and their acceptance criteria.

Story	User story	Acceptance criteria	Covers	Priority
US-06.1 Generate Study Plan Story	As a student, I need a personalized study plan built from my courses, deadlines and availability, so that I know what to study and when.	• Given courses, deadlines and availability, when the student clicks "Generate study plan", then study tasks are placed only inside the available time slots. • Given two tasks compete for one slot, when the plan is generated, then the higher-priority task gets the slot. • Given the AI returns a plan that does not match the expected structure, when it is received, then it is rejected and the message "Plan could not be generated" is shown. • Given a generated plan is rejected, when the student opens the plan page, then the previously saved plan is still shown.	FR-06.1, FR-06.2, FR-06.4, FR-06.5	High
US-06.2 Regenerate Study Plan Story	As a student, I need to regenerate my study plan after my availability or a deadline changes, so that the plan stays realistic.	• Given the availability or a deadline has changed, when the student clicks "Regenerate", then a new plan is created from the new data. • Given the new plan is rejected as invalid, when regeneration ends, then the previous plan is kept.	FR-06.3	Medium

FR-07 — Enable Smart Scheduling

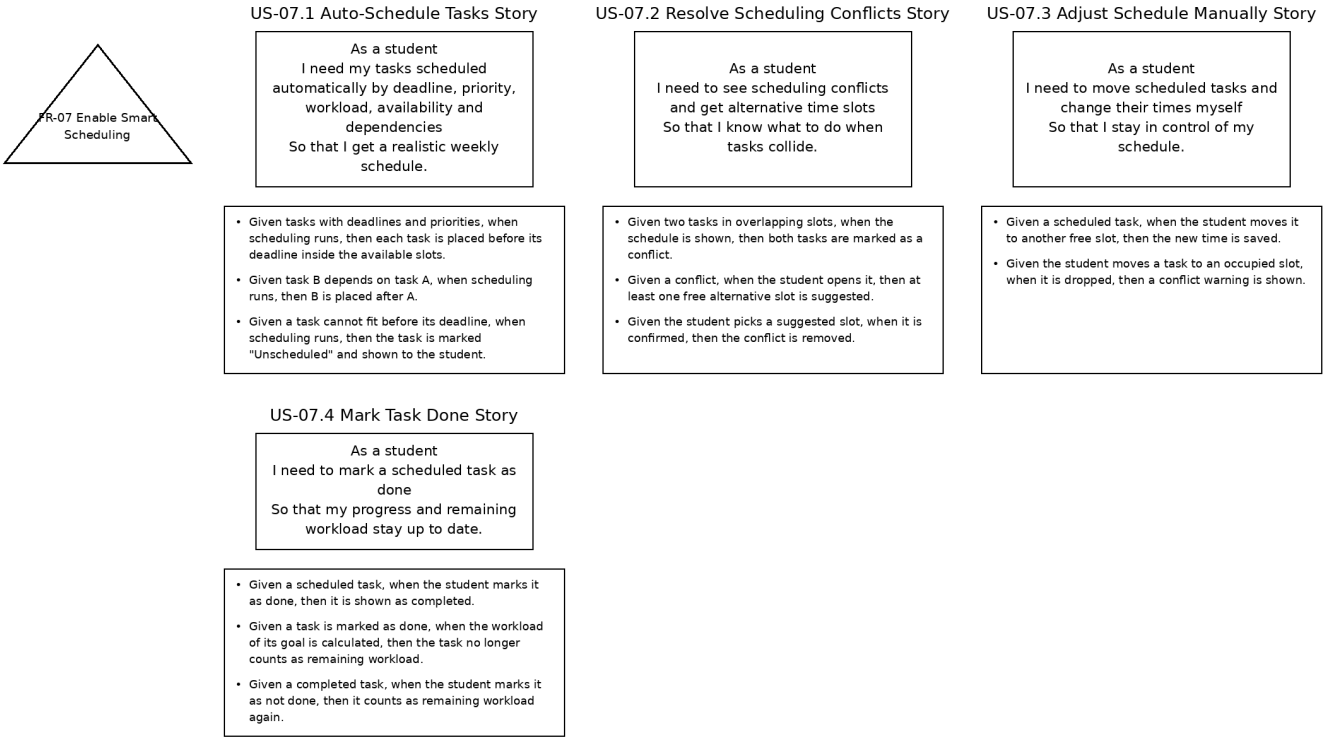


Figure 7 — Enable Smart Scheduling feature (FR-07): user stories and their acceptance criteria.

Story	User story	Acceptance criteria	Covers	Priority
US-07.1 Auto-Schedule Tasks Story	As a student, I need my tasks scheduled automatically by deadline, priority, workload, availability and dependencies, so that I get a realistic weekly schedule.	• Given tasks with deadlines and priorities, when scheduling runs, then each task is placed before its deadline inside the available slots. • Given task B depends on task A, when scheduling runs, then B is placed after A. • Given a task cannot fit before its deadline, when scheduling runs, then the task is marked "Unscheduled" and shown to the student.	FR-07.1	High
US-07.2 Resolve Scheduling Conflicts Story	As a student, I need to see scheduling conflicts and get alternative time slots, so that I know what to do when tasks collide.	• Given two tasks in overlapping slots, when the schedule is shown, then both tasks are marked as a conflict. • Given a conflict, when the student opens it, then at least one free alternative slot is suggested. • Given the student picks a suggested slot, when it is	FR-07.2, FR-07.3	High

Story	User story	Acceptance criteria	Covers	Priority
		confirmed, then the conflict is removed.		
US-07.3 Adjust Schedule Manually Story	As a student, I need to move scheduled tasks and change their times myself, so that I stay in control of my schedule.	• Given a scheduled task, when the student moves it to another free slot, then the new time is saved. • Given the student moves a task to an occupied slot, when it is dropped, then a conflict warning is shown.	FR-07.4	Medium
US-07.4 Mark Task Done Story	As a student, I need to mark a scheduled task as done, so that my progress and remaining workload stay up to date.	• Given a scheduled task, when the student marks it as done, then it is shown as completed. • Given a task is marked as done, when the workload of its goal is calculated, then the task no longer counts as remaining workload. • Given a completed task, when the student marks it as not done, then it counts as remaining workload again.	FR-07.5	High

FR-08 — Enable Adaptive Planning

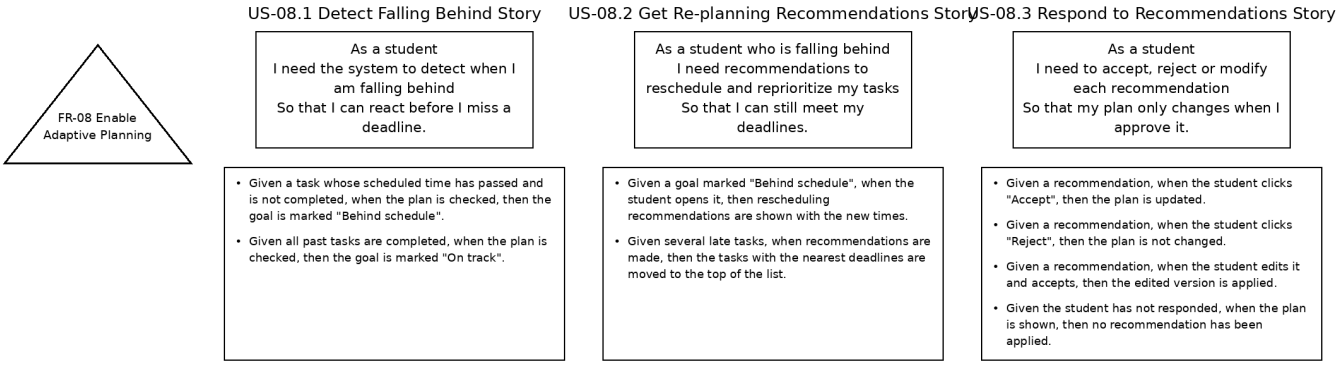


Figure 8 — Enable Adaptive Planning feature (FR-08): user stories and their acceptance criteria.

Story	User story	Acceptance criteria	Covers	Priority
US-08.1 Detect Falling Behind Story	As a student, I need the system to detect when I am falling behind, so that I can react before I miss a deadline.	• Given a task whose scheduled time has passed and is not completed, when the plan is checked, then the goal is marked "Behind schedule".	FR-08.1	High

Story	User story	Acceptance criteria	Covers	Priority
		• Given all past tasks are completed, when the plan is checked, then the goal is marked "On track".		
US-08.2 Get Re-planning Recommendations Story	As a student who is falling behind, I need recommendations to reschedule and reprioritize my tasks, so that I can still meet my deadlines.	• Given a goal marked "Behind schedule", when the student opens it, then rescheduling recommendations are shown with the new times. • Given several late tasks, when recommendations are made, then the tasks with the nearest deadlines are moved to the top of the list.	FR-08.2, FR-08.3	Medium
US-08.3 Respond to Recommendations Story	As a student, I need to accept, reject or modify each recommendation, so that my plan only changes when I approve it.	• Given a recommendation, when the student clicks "Accept", then the plan is updated. • Given a recommendation, when the student clicks "Reject", then the plan is not changed. • Given a recommendation, when the student edits it and accepts, then the edited version is applied. • Given the student has not responded, when the plan is shown, then no recommendation has been applied.	FR-08.4, FR-08.5, FR-08.6, FR-08.7	Medium

FR-09 — Manage Career Roadmaps

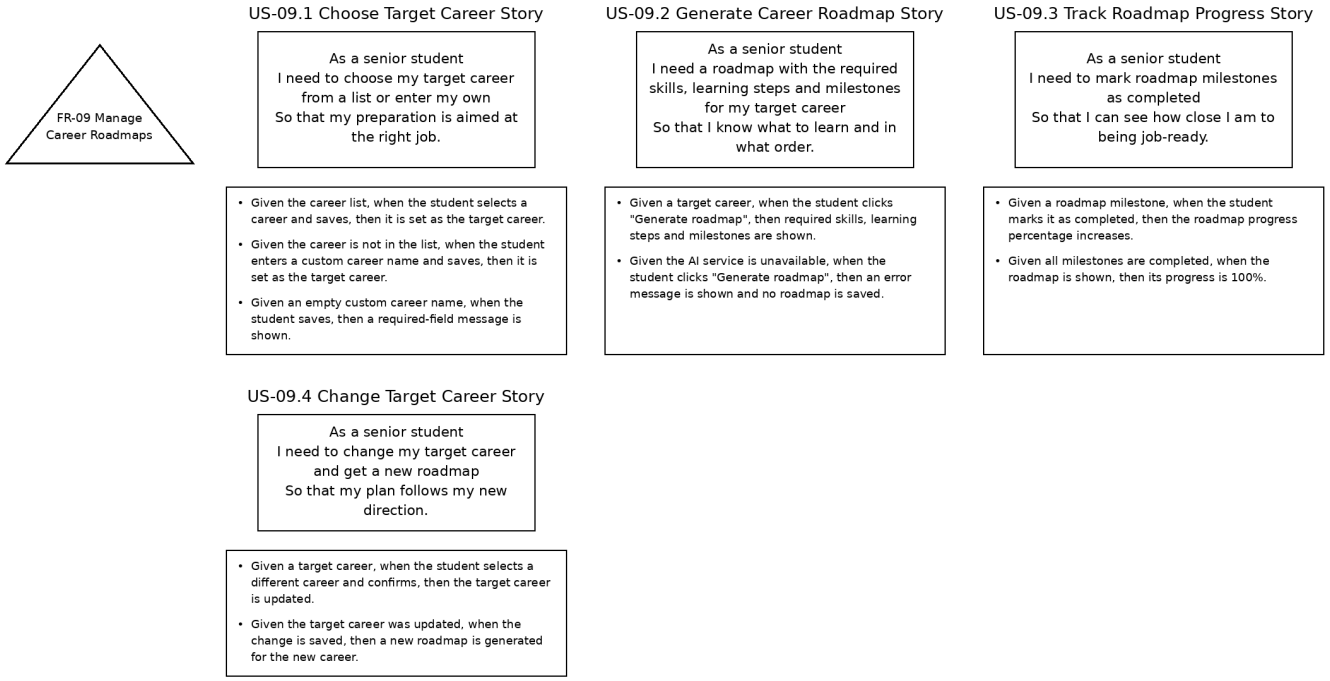


Figure 9 — Manage Career Roadmaps feature (FR-09): user stories and their acceptance criteria.

Story	User story	Acceptance criteria	Covers	Priority
US-09.1 Choose Target Career Story	As a senior student, I need to choose my target career from a list or enter my own, so that my preparation is aimed at the right job.	• Given the career list, when the student selects a career and saves, then it is set as the target career. • Given the career is not in the list, when the student enters a custom career name and saves, then it is set as the target career. • Given an empty custom career name, when the student saves, then a required-field message is shown.	FR-09.1, FR-09.2	Low
US-09.2 Generate Career Roadmap Story	As a senior student, I need a roadmap with the required skills, learning steps and milestones for my target career, so that I know what to learn and in what order.	• Given a target career, when the student clicks "Generate roadmap", then required skills, learning steps and milestones are shown. • Given the AI service is unavailable, when the student clicks "Generate roadmap", then an error message is shown and no roadmap is saved.	FR-09.3	Low

Story	User story	Acceptance criteria	Covers	Priority
US-09.3 Track Roadmap Progress Story	As a senior student, I need to mark roadmap milestones as completed, so that I can see how close I am to being job-ready.	• Given a roadmap milestone, when the student marks it as completed, then the roadmap progress percentage increases. • Given all milestones are completed, when the roadmap is shown, then its progress is 100%.	FR-09.4	Low
US-09.4 Change Target Career Story	As a senior student, I need to change my target career and get a new roadmap, so that my plan follows my new direction.	• Given a target career, when the student selects a different career and confirms, then the target career is updated. • Given the target career was updated, when the change is saved, then a new roadmap is generated for the new career.	FR-09.5, FR-09.6	Low

FR-10 — Track Skills

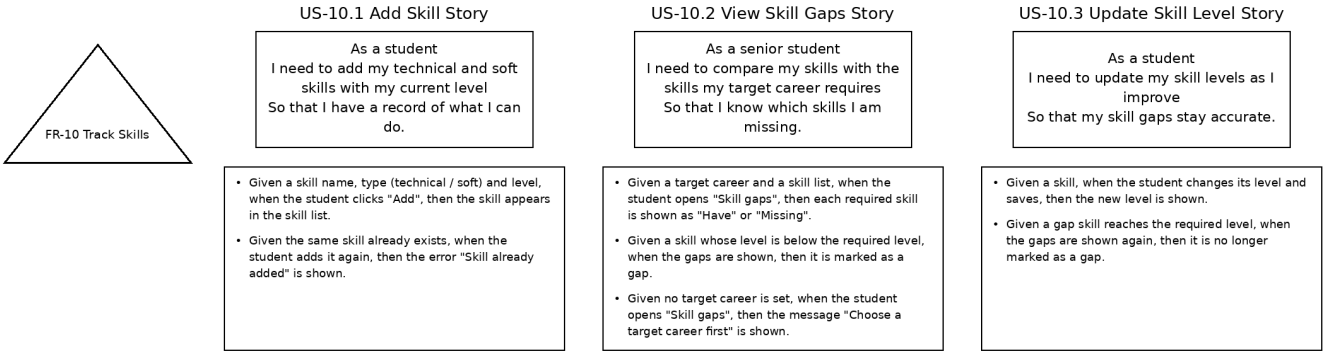


Figure 10 — Track Skills feature (FR-10): user stories and their acceptance criteria.

Story	User story	Acceptance criteria	Covers	Priority
US-10.1 Add Skill Story	As a student, I need to add my technical and soft skills with my current level, so that I have a record of what I can do.	• Given a skill name, type (technical / soft) and level, when the student clicks "Add", then the skill appears in the skill list. • Given the same skill already exists, when the student adds it again, then the error "Skill already added" is shown.	FR-10.1	Low
US-10.2 View Skill Gaps Story	As a senior student, I need to compare my skills with the skills my target career requires,	• Given a target career and a skill list, when the student opens "Skill gaps", then each required skill is	FR-10.2, FR-10.3	Low

Story	User story	Acceptance criteria	Covers	Priority
	so that I know which skills I am missing.	shown as "Have" or "Missing". • Given a skill whose level is below the required level, when the gaps are shown, then it is marked as a gap. • Given no target career is set, when the student opens "Skill gaps", then the message "Choose a target career first" is shown.		
US-10.3 Update Skill Level Story	As a student, I need to update my skill levels as I improve, so that my skill gaps stay accurate.	• Given a skill, when the student changes its level and saves, then the new level is shown. • Given a gap skill reaches the required level, when the gaps are shown again, then it is no longer marked as a gap.	FR-10.4	Low

FR-11 — Manage Projects and Portfolios

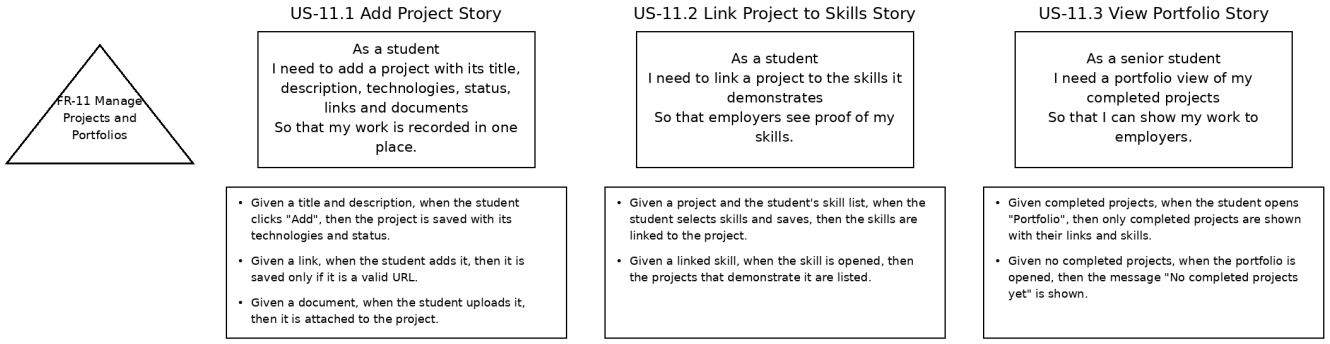


Figure 11 — Manage Projects and Portfolios feature (FR-11): user stories and their acceptance criteria.

Story	User story	Acceptance criteria	Covers	Priority
US-11.1 Add Project Story	As a student, I need to add a project with its title, description, technologies, status, links and documents, so that my work is recorded in one place.	• Given a title and description, when the student clicks "Add", then the project is saved with its technologies and status. • Given a link, when the student adds it, then it is saved only if it is a valid URL. • Given a document, when the student uploads it, then it is attached to the project.	FR-11.1, FR-11.4	Low
US-11.2 Link Project	As a student, I need to link a project to the skills it	• Given a project and the student's skill list, when the student selects skills and	FR-11.2	Low

Story	User story	Acceptance criteria	Covers	Priority
to Skills Story	demonstrates, so that employers see proof of my skills.	saves, then the skills are linked to the project. • Given a linked skill, when the skill is opened, then the projects that demonstrate it are listed.		
US-11.3 View Portfolio Story	As a senior student, I need a portfolio view of my completed projects, so that I can show my work to employers.	• Given completed projects, when the student opens "Portfolio", then only completed projects are shown with their links and skills. • Given no completed projects, when the portfolio is opened, then the message "No completed projects yet" is shown.	FR-11.3	Low

FR-12 — Manage Internship and Job Preparation

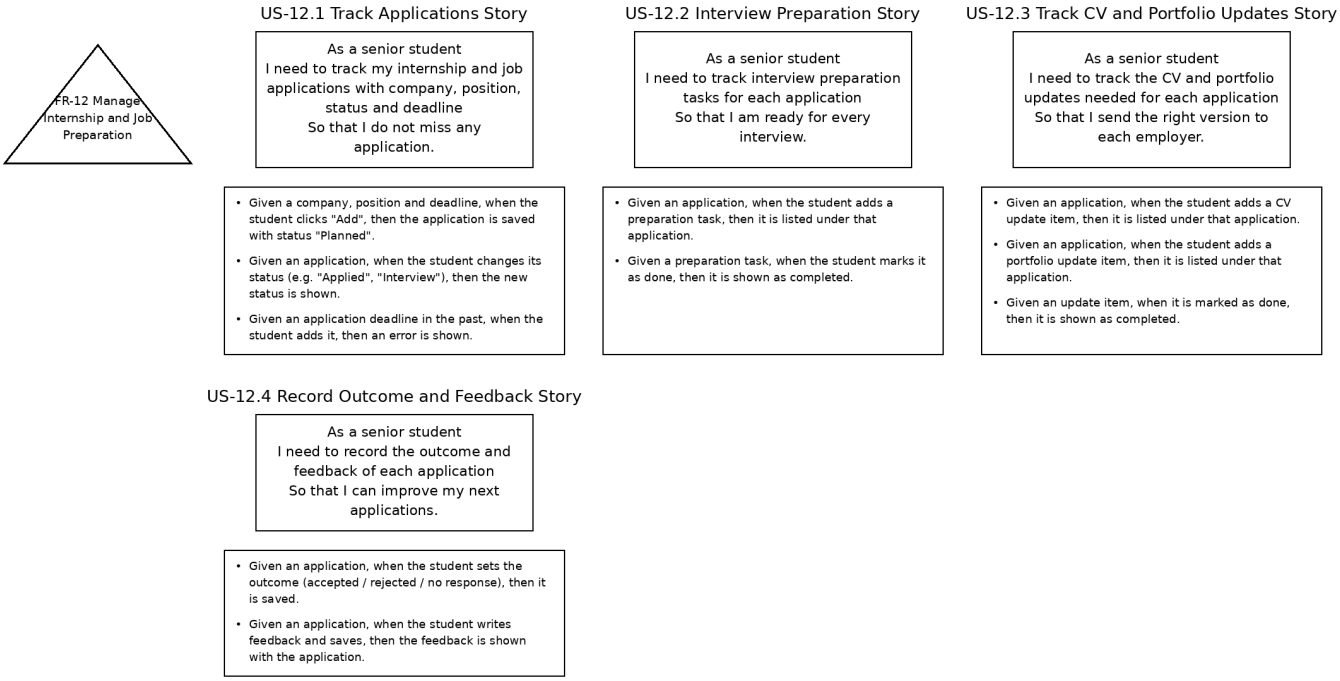


Figure 12 — Manage Internship and Job Preparation feature (FR-12): user stories and their acceptance criteria.

Story	User story	Acceptance criteria	Covers	Priority
US-12.1 Track Applications Story	As a senior student, I need to track my internship and job applications with company, position, status and deadline, so that I do not miss any application.	• Given a company, position and deadline, when the student clicks "Add", then the application is saved with status "Planned". • Given an application,	FR-12.1	Low

Story	User story	Acceptance criteria	Covers	Priority
		when the student changes its status (e.g. "Applied", "Interview"), then the new status is shown. • Given an application deadline in the past, when the student adds it, then an error is shown.		
US-12.2 Interview Preparation Story	As a senior student, I need to track interview preparation tasks for each application, so that I am ready for every interview.	• Given an application, when the student adds a preparation task, then it is listed under that application. • Given a preparation task, when the student marks it as done, then it is shown as completed.	FR-12.2	Low
US-12.3 Track CV and Portfolio Updates Story	As a senior student, I need to track the CV and portfolio updates needed for each application, so that I send the right version to each employer.	• Given an application, when the student adds a CV update item, then it is listed under that application. • Given an application, when the student adds a portfolio update item, then it is listed under that application. • Given an update item, when it is marked as done, then it is shown as completed.	FR-12.3, FR-12.4	Low
US-12.4 Record Outcome and Feedback Story	As a senior student, I need to record the outcome and feedback of each application, so that I can improve my next applications.	• Given an application, when the student sets the outcome (accepted / rejected / no response), then it is saved. • Given an application, when the student writes feedback and saves, then the feedback is shown with the application.	FR-12.5, FR-12.6	Low

FR-13 — Manage Habits

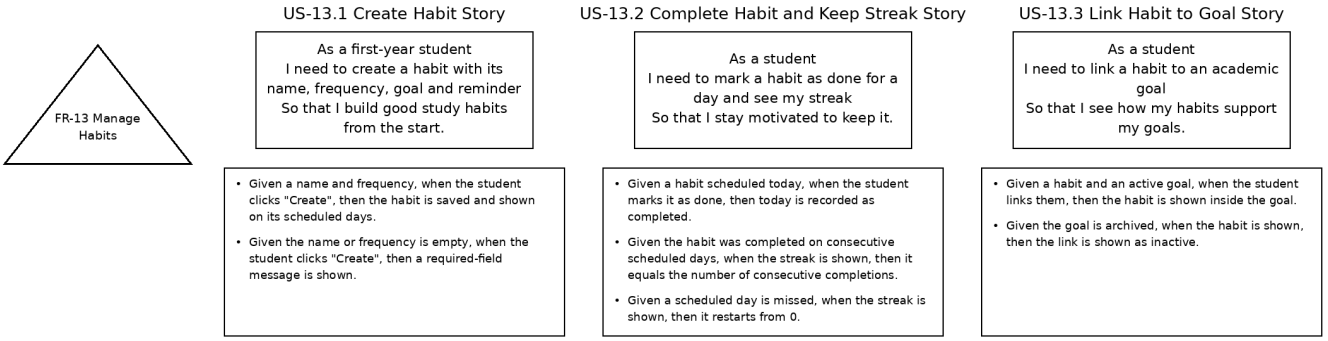


Figure 13 — Manage Habits feature (FR-13): user stories and their acceptance criteria.

Story	User story	Acceptance criteria	Covers	Priority
US-13.1 Create Habit Story	As a first-year student, I need to create a habit with its name, frequency, goal and reminder, so that I build good study habits from the start.	• Given a name and frequency, when the student clicks "Create", then the habit is saved and shown on its scheduled days. • Given the name or frequency is empty, when the student clicks "Create", then a required-field message is shown.	FR-13.1	High
US-13.2 Complete Habit and Keep Streak Story	As a student, I need to mark a habit as done for a day and see my streak, so that I stay motivated to keep it.	• Given a habit scheduled today, when the student marks it as done, then today is recorded as completed. • Given the habit was completed on consecutive scheduled days, when the streak is shown, then it equals the number of consecutive completions. • Given a scheduled day is missed, when the streak is shown, then it restarts from 0.	FR-13.2, FR-13.3	High
US-13.3 Link Habit to Goal Story	As a student, I need to link a habit to an academic goal, so that I see how my habits support my goals.	• Given a habit and an active goal, when the student links them, then the habit is shown inside the goal. • Given the goal is archived, when the habit is shown, then the link is shown as inactive.	FR-13.4	Medium

FR-14 — Analyze Study Workload

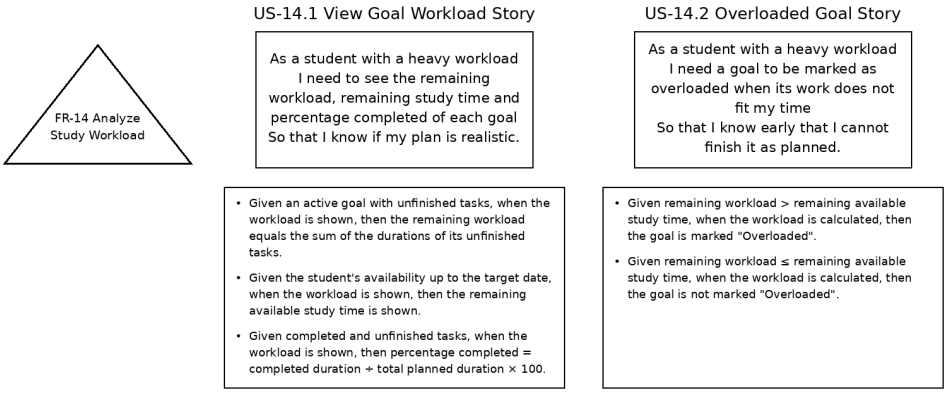


Figure 14 — Analyze Study Workload feature (FR-14): user stories and their acceptance criteria.

Story	User story	Acceptance criteria	Covers	Priority
US-14.1 View Goal Workload Story	As a student with a heavy workload, I need to see the remaining workload, remaining study time and percentage completed of each goal, so that I know if my plan is realistic.	• Given an active goal with unfinished tasks, when the workload is shown, then the remaining workload equals the sum of the durations of its unfinished tasks. • Given the student's availability up to the target date, when the workload is shown, then the remaining available study time is shown. • Given completed and unfinished tasks, when the workload is shown, then percentage completed = completed duration ÷ total planned duration × 100.	FR-14.1, FR-14.2, FR-14.3, FR-14.4	Medium
US-14.2 Overloaded Goal Story	As a student with a heavy workload, I need a goal to be marked as overloaded when its work does not fit my time, so that I know early that I cannot finish it as planned.	• Given remaining workload > remaining available study time, when the workload is calculated, then the goal is marked "Overloaded". • Given remaining workload ≤ remaining available study time, when the workload is calculated, then the goal is not marked "Overloaded".	FR-14.5	Medium

FR-15 — Predict and Warn Early

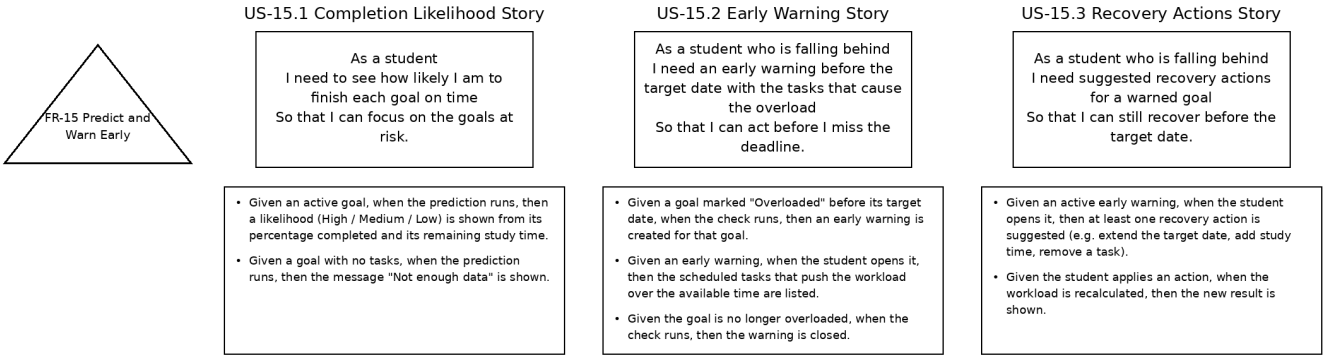


Figure 15 — Predict and Warn Early feature (FR-15): user stories and their acceptance criteria.

Story	User story	Acceptance criteria	Covers	Priority
US-15.1 Completion Likelihood Story	As a student, I need to see how likely I am to finish each goal on time, so that I can focus on the goals at risk.	• Given an active goal, when the prediction runs, then a likelihood (High / Medium / Low) is shown from its percentage completed and its remaining study time. • Given a goal with no tasks, when the prediction runs, then the message "Not enough data" is shown.	FR-15.1	Medium
US-15.2 Early Warning Story	As a student who is falling behind, I need an early warning before the target date with the tasks that cause the overload, so that I can act before I miss the deadline.	• Given a goal marked "Overloaded" before its target date, when the check runs, then an early warning is created for that goal. • Given an early warning, when the student opens it, then the scheduled tasks that push the workload over the available time are listed. • Given the goal is no longer overloaded, when the check runs, then the warning is closed.	FR-15.2, FR-15.3	Medium
US-15.3 Recovery Actions Story	As a student who is falling behind, I need suggested recovery actions for a warned goal, so that I can still recover before the target date.	• Given an active early warning, when the student opens it, then at least one recovery action is suggested (e.g. extend the target date, add study time, remove a task). • Given the student applies an action, when the workload is recalculated, then the new result is shown.	FR-15.4	Medium

FR-16 — Coach Students with AI

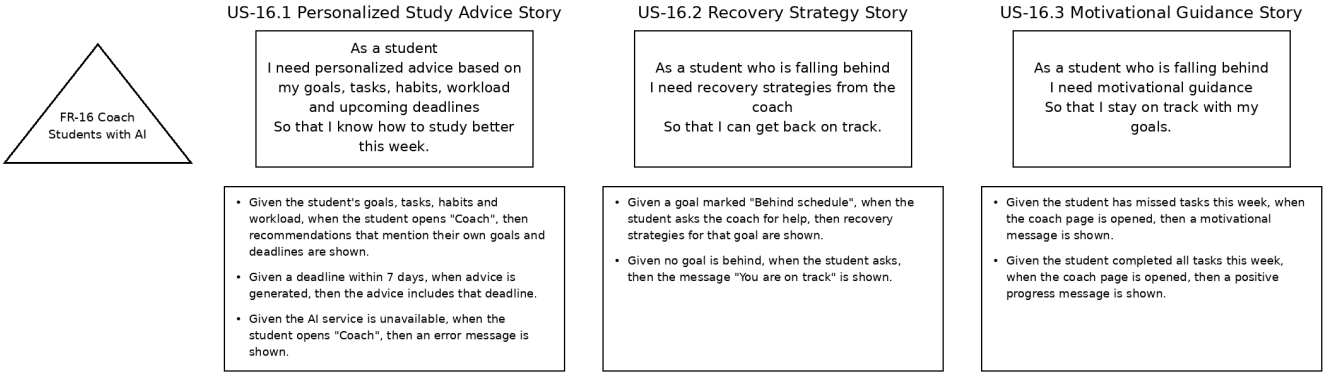


Figure 16 — Coach Students with AI feature (FR-16): user stories and their acceptance criteria.

Story	User story	Acceptance criteria	Covers	Priority
US-16.1 Personalized Study Advice Story	As a student, I need personalized advice based on my goals, tasks, habits, workload and upcoming deadlines, so that I know how to study better this week.	• Given the student's goals, tasks, habits and workload, when the student opens "Coach", then recommendations that mention their own goals and deadlines are shown. • Given a deadline within 7 days, when advice is generated, then the advice includes that deadline. • Given the AI service is unavailable, when the student opens "Coach", then an error message is shown.	FR-16.1, FR-16.2	Low
US-16.2 Recovery Strategy Story	As a student who is falling behind, I need recovery strategies from the coach, so that I can get back on track.	• Given a goal marked "Behind schedule", when the student asks the coach for help, then recovery strategies for that goal are shown. • Given no goal is behind, when the student asks, then the message "You are on track" is shown.	FR-16.3	Low
US-16.3 Motivational Guidance Story	As a student who is falling behind, I need motivational guidance, so that I stay on track with my goals.	• Given the student has missed tasks this week, when the coach page is opened, then a motivational message is shown. • Given the student completed all tasks this week, when the coach page is opened, then a positive progress message is shown.	FR-16.4	Low

FR-17 — Display Student Dashboard

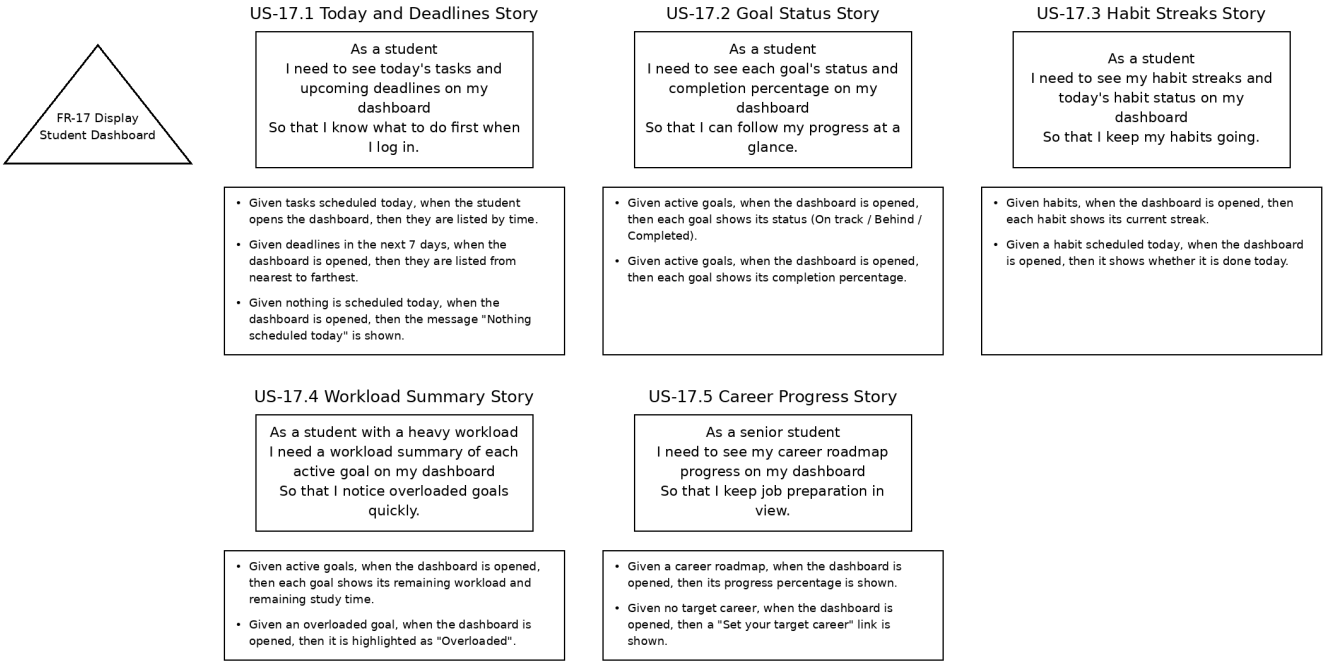


Figure 17 — Display Student Dashboard feature (FR-17): user stories and their acceptance criteria.

Story	User story	Acceptance criteria	Covers	Priority
US-17.1 Today and Deadlines Story	As a student, I need to see today's tasks and upcoming deadlines on my dashboard, so that I know what to do first when I log in.	• Given tasks scheduled today, when the student opens the dashboard, then they are listed by time. • Given deadlines in the next 7 days, when the dashboard is opened, then they are listed from nearest to farthest. • Given nothing is scheduled today, when the dashboard is opened, then the message "Nothing scheduled today" is shown.	FR-17.1	High
US-17.2 Goal Status Story	As a student, I need to see each goal's status and completion percentage on my dashboard, so that I can follow my progress at a glance.	• Given active goals, when the dashboard is opened, then each goal shows its status (On track / Behind / Completed). • Given active goals, when the dashboard is opened, then each goal shows its completion percentage.	FR-17.2, FR-17.3	High
US-17.3 Habit Streaks Story	As a student, I need to see my habit streaks and today's habit status on my dashboard, so that I keep my habits going.	• Given habits, when the dashboard is opened, then each habit shows its current streak. • Given a habit scheduled	FR-17.4	High

Story	User story	Acceptance criteria	Covers	Priority
		today, when the dashboard is opened, then it shows whether it is done today.		
US-17.4 Workload Summary Story	As a student with a heavy workload, I need a workload summary of each active goal on my dashboard, so that I notice overloaded goals quickly.	• Given active goals, when the dashboard is opened, then each goal shows its remaining workload and remaining study time. • Given an overloaded goal, when the dashboard is opened, then it is highlighted as "Overloaded".	FR-17.5	Medium
US-17.5 Career Progress Story	As a senior student, I need to see my career roadmap progress on my dashboard, so that I keep job preparation in view.	• Given a career roadmap, when the dashboard is opened, then its progress percentage is shown. • Given no target career, when the dashboard is opened, then a "Set your target career" link is shown.	FR-17.6	Low

FR-18 — Send Notifications

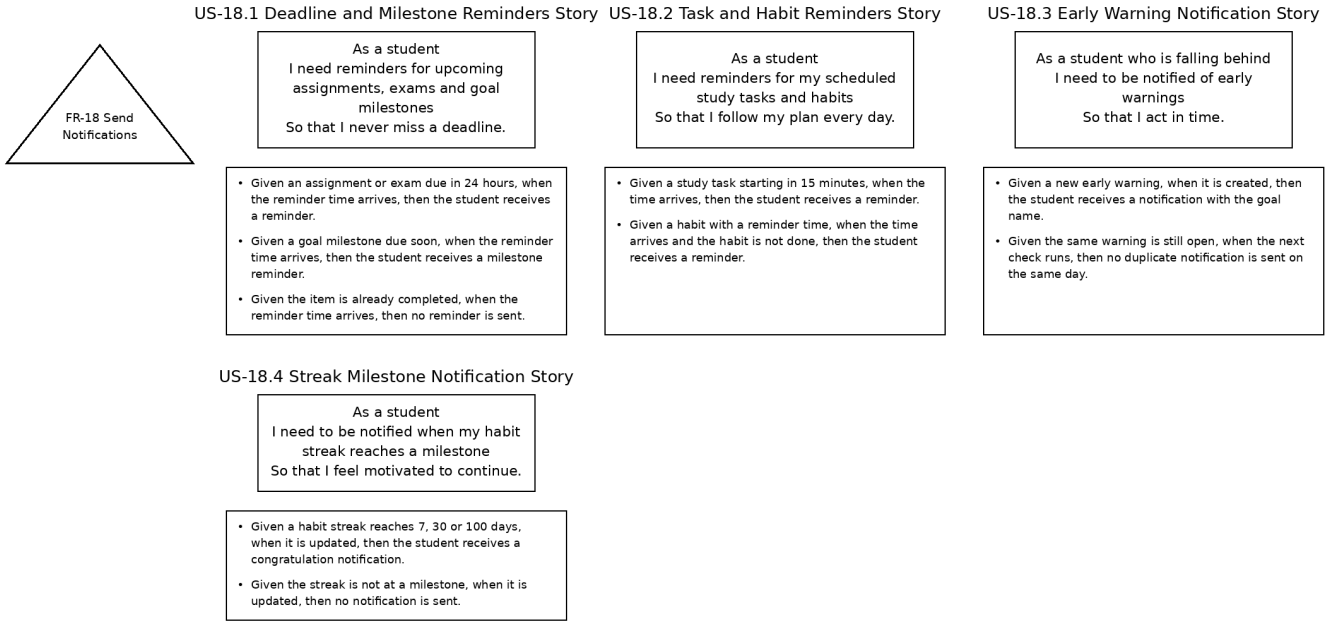


Figure 18 — Send Notifications feature (FR-18): user stories and their acceptance criteria.

Story	User story	Acceptance criteria	Covers	Priority
US-18.1 Deadline and Milestone Reminders Story	As a student, I need reminders for upcoming assignments, exams and goal milestones, so that I never miss a deadline.	• Given an assignment or exam due in 24 hours, when the reminder time arrives, then the student receives a reminder. • Given a goal milestone due soon, when the reminder time arrives, then the student receives a milestone reminder. • Given the item is already completed, when the reminder time arrives, then no reminder is sent.	FR-18.1, FR-18.5	High
US-18.2 Task and Habit Reminders Story	As a student, I need reminders for my scheduled study tasks and habits, so that I follow my plan every day.	• Given a study task starting in 15 minutes, when the time arrives, then the student receives a reminder. • Given a habit with a reminder time, when the time arrives and the habit is not done, then the student receives a reminder.	FR-18.2, FR-18.3	High
US-18.3 Early Warning Notification Story	As a student who is falling behind, I need to be notified of early warnings, so that I act in time.	• Given a new early warning, when it is created, then the student receives a notification with the goal name. • Given the same warning is still open, when the next check runs, then no duplicate notification is sent on the same day.	FR-18.4	Medium
US-18.4 Streak Milestone Notification Story	As a student, I need to be notified when my habit streak reaches a milestone, so that I feel motivated to continue.	• Given a habit streak reaches 7, 30 or 100 days, when it is updated, then the student receives a congratulation notification. • Given the streak is not at a milestone, when it is updated, then no notification is sent.	FR-18.6	Medium

FR-19 — Manage Study Groups and Accountability

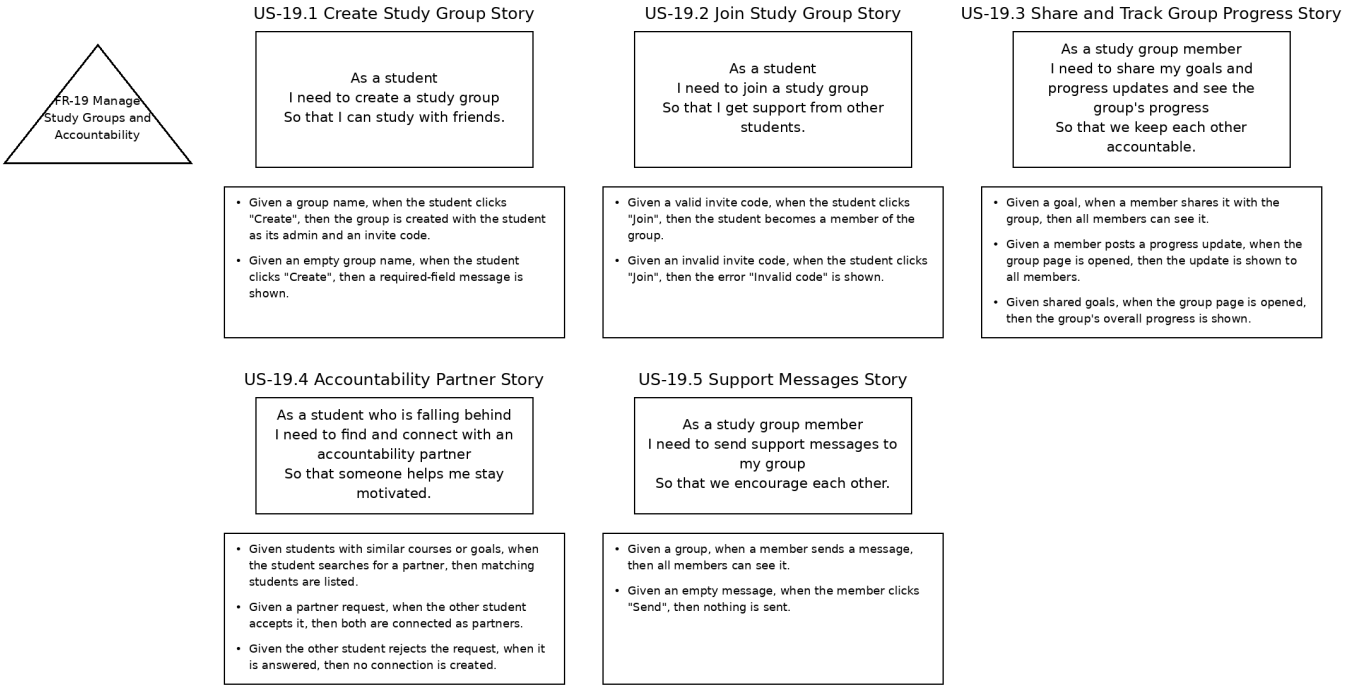


Figure 19 — Manage Study Groups and Accountability feature (FR-19): user stories and their acceptance criteria.

Story	User story	Acceptance criteria	Covers	Priority
US-19.1 Create Study Group Story	As a student, I need to create a study group, so that I can study with friends.	• Given a group name, when the student clicks "Create", then the group is created with the student as its admin and an invite code. • Given an empty group name, when the student clicks "Create", then a required-field message is shown.	FR-19.1	Low
US-19.2 Join Study Group Story	As a student, I need to join a study group, so that I get support from other students.	• Given a valid invite code, when the student clicks "Join", then the student becomes a member of the group. • Given an invalid invite code, when the student clicks "Join", then the error "Invalid code" is shown.	FR-19.2	Low
US-19.3 Share and Track Group Progress Story	As a study group member, I need to share my goals and progress updates and see the group's progress, so that we keep each other accountable.	• Given a goal, when a member shares it with the group, then all members can see it. • Given a member posts a	FR-19.3, FR-19.4, FR-19.7	Low

Story	User story	Acceptance criteria	Covers	Priority
		progress update, when the group page is opened, then the update is shown to all members. • Given shared goals, when the group page is opened, then the group's overall progress is shown.		
US-19.4 Accountability Partner Story	As a student who is falling behind, I need to find and connect with an accountability partner, so that someone helps me stay motivated.	• Given students with similar courses or goals, when the student searches for a partner, then matching students are listed. • Given a partner request, when the other student accepts it, then both are connected as partners. • Given the other student rejects the request, when it is answered, then no connection is created.	FR-19.5, FR-19.6	Low
US-19.5 Support Messages Story	As a study group member, I need to send support messages to my group, so that we encourage each other.	• Given a group, when a member sends a message, then all members can see it. • Given an empty message, when the member clicks "Send", then nothing is sent.	FR-19.8	Low

FR-20 — Manage Administration

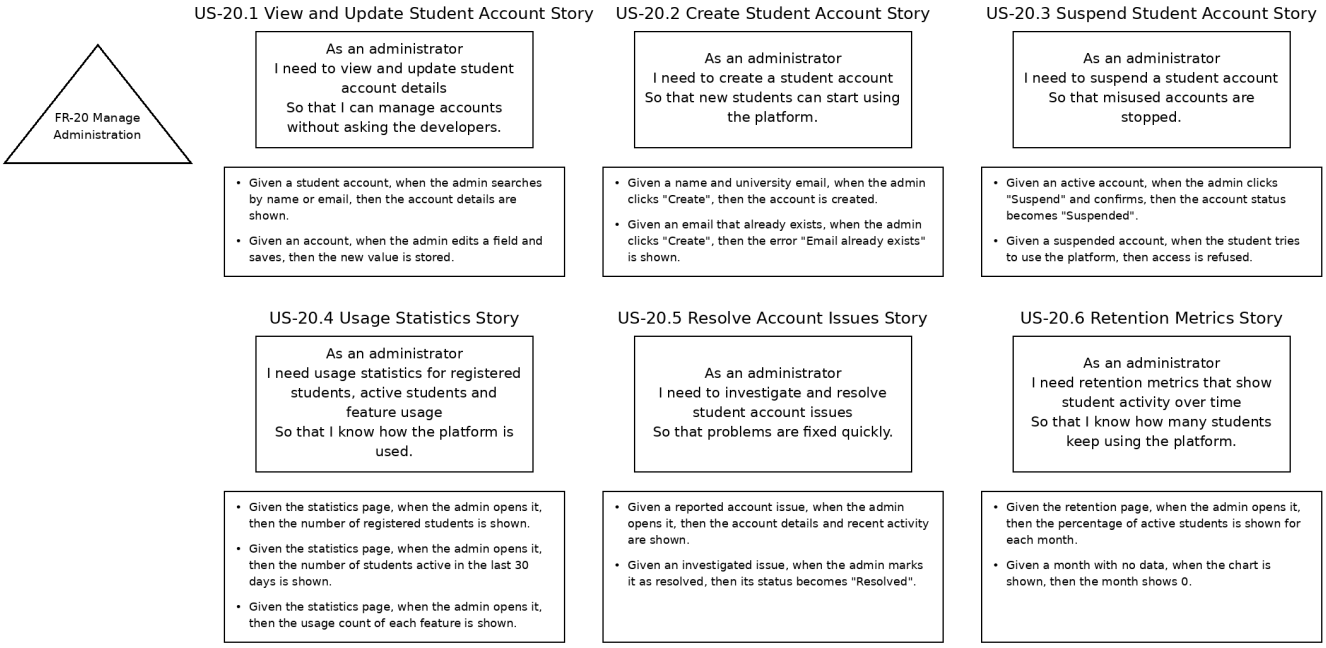


Figure 20 — Manage Administration feature (FR-20): user stories and their acceptance criteria.

Story	User story	Acceptance criteria	Covers	Priority
US-20.1 View and Update Student Account Story	As an administrator, I need to view and update student account details, so that I can manage accounts without asking the developers.	• Given a student account, when the admin searches by name or email, then the account details are shown. • Given an account, when the admin edits a field and saves, then the new value is stored.	FR-20.1, FR-20.3	Medium
US-20.2 Create Student Account Story	As an administrator, I need to create a student account, so that new students can start using the platform.	• Given a name and university email, when the admin clicks "Create", then the account is created. • Given an email that already exists, when the admin clicks "Create", then the error "Email already exists" is shown.	FR-20.2	Medium
US-20.3 Suspend Student Account Story	As an administrator, I need to suspend a student account, so that misused accounts are stopped.	• Given an active account, when the admin clicks "Suspend" and confirms, then the account status becomes "Suspended". • Given a suspended account, when the student tries to use the platform, then access is refused.	FR-20.4	Medium

Story	User story	Acceptance criteria	Covers	Priority
US-20.4 Usage Statistics Story	As an administrator, I need usage statistics for registered students, active students and feature usage, so that I know how the platform is used.	• Given the statistics page, when the admin opens it, then the number of registered students is shown. • Given the statistics page, when the admin opens it, then the number of students active in the last 30 days is shown. • Given the statistics page, when the admin opens it, then the usage count of each feature is shown.	FR-20.5, FR-20.6, FR-20.7	Medium
US-20.5 Resolve Account Issues Story	As an administrator, I need to investigate and resolve student account issues, so that problems are fixed quickly.	• Given a reported account issue, when the admin opens it, then the account details and recent activity are shown. • Given an investigated issue, when the admin marks it as resolved, then its status becomes "Resolved".	FR-20.8, FR-20.9	Medium
US-20.6 Retention Metrics Story	As an administrator, I need retention metrics that show student activity over time, so that I know how many students keep using the platform.	• Given the retention page, when the admin opens it, then the percentage of active students is shown for each month. • Given a month with no data, when the chart is shown, then the month shows 0.	FR-20.10	Low

C9 — Estimation & Sprint Plan

Given (team decisions): 1 story point = 1 working day. 1 week = 5 working days. Sprint length = 2 weeks (10 working days). Deadline = 12 weeks. Team = 5 developers.

Planning Poker results

Feature	User story	Points	Why this size?
FR-01 Manage Student Profile	US-01.1 Create Profile	3	Profile form with availability slots + required-field and time-slot validation

Feature	User story	Points	Why this size?
FR-01 Manage Student Profile	US-01.2 Update Profile	2	Re-uses the create form; update + validation
FR-01 Manage Student Profile	US-01.3 View Profile	1	Read-only profile page
FR-02 Manage Academic Goals	US-02.1 Create Goal	3	Goal form + validation (required fields, future date)
FR-02 Manage Academic Goals	US-02.2 Edit Goal	2	Edit existing goal + date validation
FR-02 Manage Academic Goals	US-02.3 Complete Goal	1	Status change to Completed
FR-02 Manage Academic Goals	US-02.4 Archive Goal	2	Archive + restore (two status changes and a filtered list)
FR-02 Manage Academic Goals	US-02.5 Delete Goal	1	Delete with confirmation
FR-03 Manage Courses	US-03.1 Add Course	3	Course form + credits/workload + duplicate-code check
FR-03 Manage Courses	US-03.2 View Course Progress	3	Joins course with its assignments, completed vs outstanding
FR-03 Manage Courses	US-03.3 Update Course	2	Edit course + duplicate-code check
FR-03 Manage Courses	US-03.4 Remove Course	2	Remove course + cascade warning for its assignments
FR-04 Manage Assignments and Exams	US-04.1 Add Assignment	3	Assignment form + validation + default status
FR-04 Manage Assignments and Exams	US-04.2 Add Exam	2	Exam form re-uses the assignment form pattern
FR-04 Manage Assignments and Exams	US-04.3 Track Assignment Status	2	Status change that also updates course progress
FR-04 Manage Assignments and Exams	US-04.4 Exam Preparation Reminder	2	Reminder setting saved with the exam + date check
FR-05 Plan Goals with AI	US-05.1 Generate Goal Plan	8	First AI integration: prompt, call, parse milestones/tasks, AI-unavailable handling

Feature	User story	Points	Why this size?
FR-05 Plan Goals with AI	US-05.2 Review and Edit AI Plan	5	Editable review screen for both milestones and tasks, nothing saved before approval
FR-05 Plan Goals with AI	US-05.3 Approve AI Plan	2	Approve saves the reviewed plan; discard drops it
FR-06 Generate Study Plans	US-06.1 Generate Study Plan	8	AI plan within availability + priority placement + schema validation + keep previous plan (re-uses AI client from US-05.1)
FR-06 Generate Study Plans	US-06.2 Regenerate Study Plan	3	Re-runs generation with new data; keeps old plan on failure
FR-07 Enable Smart Scheduling	US-07.1 Auto-Schedule Tasks	8	Scheduling algorithm: deadlines, priorities, availability and task dependencies
FR-07 Enable Smart Scheduling	US-07.2 Resolve Scheduling Conflicts	5	Overlap detection + finding free alternative slots
FR-07 Enable Smart Scheduling	US-07.3 Adjust Schedule Manually	3	Move a task to a new slot + conflict warning
FR-07 Enable Smart Scheduling	US-07.4 Mark Task Done	2	Status change on a scheduled task that updates progress and remaining workload
FR-08 Enable Adaptive Planning	US-08.1 Detect Falling Behind	3	Rule: past, unfinished tasks → Behind schedule
FR-08 Enable Adaptive Planning	US-08.2 Get Re-planning Recommendations	5	Builds reschedule/reprioritize suggestions from the schedule
FR-08 Enable Adaptive Planning	US-08.3 Respond to Recommendations	3	Accept / reject / modify flow with approval before applying
FR-09 Manage Career Roadmaps	US-09.1 Choose Target Career	2	Select from a list or enter a custom career + validation
FR-09 Manage Career Roadmaps	US-09.2 Generate Career Roadmap	5	AI roadmap generation (re-uses the AI client) + error handling
FR-09 Manage Career Roadmaps	US-09.3 Track Roadmap Progress	2	Mark milestones done + progress percentage
FR-09 Manage Career Roadmaps	US-09.4 Change Target Career	3	Change career + trigger roadmap regeneration
FR-10 Track Skills	US-10.1 Add Skill	2	Skill form + duplicate check
FR-10 Track Skills	US-10.2 View Skill Gaps	5	Compare skills and levels with the roadmap's required skills

Feature	User story	Points	Why this size?
FR-10 Track Skills	US-10.3 Update Skill Level	1	Update one field; gaps re-use US-10.2
FR-11 Manage Projects and Portfolios	US-11.1 Add Project	5	Project form + URL validation + document upload
FR-11 Manage Projects and Portfolios	US-11.2 Link Project to Skills	3	Many-to-many link between projects and skills, shown both ways
FR-11 Manage Projects and Portfolios	US-11.3 View Portfolio	3	Portfolio page: completed projects with links and skills
FR-12 Manage Internship and Job Preparation	US-12.1 Track Applications	3	Application form + status changes + deadline check
FR-12 Manage Internship and Job Preparation	US-12.2 Interview Preparation	2	Checklist of preparation tasks per application
FR-12 Manage Internship and Job Preparation	US-12.3 Track CV and Portfolio Updates	3	Two item lists (CV and portfolio updates) per application
FR-12 Manage Internship and Job Preparation	US-12.4 Record Outcome and Feedback	2	Outcome field + feedback text
FR-13 Manage Habits	US-13.1 Create Habit	3	Habit form with frequency and reminder + validation
FR-13 Manage Habits	US-13.2 Complete Habit and Keep Streak	3	Daily completion + streak calculation and reset
FR-13 Manage Habits	US-13.3 Link Habit to Goal	2	Link habit to goal + inactive state when the goal is archived
FR-14 Analyze Study Workload	US-14.1 View Goal Workload	5	Three calculations (remaining work, available time, % completed) + display
FR-14 Analyze Study Workload	US-14.2 Overloaded Goal	2	One comparison rule on top of US-14.1
FR-15 Predict and Warn Early	US-15.1 Completion Likelihood	5	Likelihood rule from % completed and remaining time + no-data case
FR-15 Predict and Warn Early	US-15.2 Early Warning	5	Creates/closes warnings and finds the tasks that cause the overload

Feature	User story	Points	Why this size?
FR-15 Predict and Warn Early	US-15.3 Recovery Actions	3	Suggests actions and recalculates after one is applied
FR-16 Coach Students with AI	US-16.1 Personalized Study Advice	5	AI advice from goals, tasks, habits and workload (re-uses AI client)
FR-16 Coach Students with AI	US-16.2 Recovery Strategy	3	Coach response for goals that are behind
FR-16 Coach Students with AI	US-16.3 Motivational Guidance	2	Message chosen from the week's completion
FR-17 Display Student Dashboard	US-17.1 Today and Deadlines	3	Dashboard layout + today's tasks + 7-day deadlines
FR-17 Display Student Dashboard	US-17.2 Goal Status	2	Goal status and % widget (data from goals)
FR-17 Display Student Dashboard	US-17.3 Habit Streaks	2	Habit streak widget (data from habits)
FR-17 Display Student Dashboard	US-17.4 Workload Summary	2	Workload widget (data from US-14.1) + overload highlight
FR-17 Display Student Dashboard	US-17.5 Career Progress	1	One progress value from the career roadmap
FR-18 Send Notifications	US-18.1 Deadline and Milestone Reminders	5	Reminder scheduler + email gateway + skip completed items
FR-18 Send Notifications	US-18.2 Task and Habit Reminders	3	Two more reminder types on the scheduler from US-18.1
FR-18 Send Notifications	US-18.3 Early Warning Notification	2	Notification on new warning + no duplicates per day
FR-18 Send Notifications	US-18.4 Streak Milestone Notification	2	Notification on streak milestones (7/30/100)
FR-19 Manage Study Groups and Accountability	US-19.1 Create Study Group	2	Create group + invite code
FR-19 Manage Study Groups and Accountability	US-19.2 Join Study Group	2	Join by invite code + invalid-code error
FR-19 Manage Study Groups and Accountability	US-19.3 Share and Track Group Progress	5	Share goals, post updates and group progress for all members

Feature	User story	Points	Why this size?
FR-19 Manage Study Groups and Accountability	US-19.4 Accountability Partner	5	Partner matching + request / accept / reject flow
FR-19 Manage Study Groups and Accountability	US-19.5 Support Messages	3	Group message list + send
FR-20 Manage Administration	US-20.1 View and Update Student Account	3	Search accounts + view and edit details
FR-20 Manage Administration	US-20.2 Create Student Account	2	Create account + duplicate-email check
FR-20 Manage Administration	US-20.3 Suspend Student Account	2	Suspend + refuse access to suspended accounts
FR-20 Manage Administration	US-20.4 Usage Statistics	5	Three statistics queries + statistics page
FR-20 Manage Administration	US-20.5 Resolve Account Issues	3	Issue view with account activity + resolve status
FR-20 Manage Administration	US-20.6 Retention Metrics	3	Monthly retention calculation + chart
Total	**73 stories**	**225**	

Computed values

1. Total points

```
Total points = Σ story points
             = 6 + 9 + 10 + 9 + 15 + 11 + 18 + 11 + 12 + 8 + 11 + 10 + 8 + 7 + 13
+ 10 + 10 + 12 + 17 + 18   (FR-01 … FR-20)
             = 225 points
```

2. Effort (person-days)

```
Effort = total points × days per point
       = 225 × 1
       = 225 person-days
```

3. Effort (person-weeks)

```
Effort = person-days ÷ 5
       = 225 ÷ 5
       = 45 person-weeks
```

4. Team size

```
Team size = effort (person-weeks) ÷ deadline (weeks) → round up
          = 45 ÷ 12
          = 3.75 → 4 developers
```

5. Capacity check

```
Capacity = team size × weeks ≥ required effort
         = 4 × 12 = 48 person-weeks ≥ 45 ✓
Why not 3? 3 × 12 = 36 < 45 ✗ — the deadline would be missed.
Our team (5): 5 × 12 = 60 person-weeks ≥ 45 ✓
```

6. Sprint capacity

```
Sprint capacity = developers × working days per sprint ÷ days per point
                = 5 × 10 ÷ 1
                = 50 points per sprint
```

7. Actual velocity

```
Actual velocity = completed story points per sprint
No sprint is finished yet → planned velocity = sprint capacity
                = 50 points per sprint (actual velocity reported after every
sprint, from PM1)
```

8. Schedule (weeks)

```
Schedule = effort (person-weeks) ÷ team size → round up
         = 45 ÷ 5
         = 9 → 9 weeks
```

9. Number of sprints

```
Number of sprints = total points ÷ sprint capacity → round up
                  = 225 ÷ 50
```

```
                    = 4.5 → 5 sprints = 10 weeks ≤ 12 weeks ✓
```

10. Max scope (points)

```
Max scope = team size × weeks × 5 ÷ days per point
          = 5 × 12 × 5 ÷ 1
          = 300 points ≥ 225 ✓ (no story has to be removed)
```

Sprint plan (by priority)

Sprint	Weeks	User stories	Points	Fits? (≤ 50)
Sprint 1	1–2	US-01.1 Create Profile (3), US-01.3 View Profile (1), US-01.2 Update Profile (2), US-02.1 Create Goal (3), US-02.2 Edit Goal (2), US-02.3 Complete Goal (1), US-03.1 Add Course (3), US-04.1 Add Assignment (3), US-04.2 Add Exam (2), US-04.3 Track Assignment Status (2), US-03.2 View Course Progress (3), US-13.1 Create Habit (3), US-13.2 Complete Habit and Keep Streak (3), US-05.1 Generate Goal Plan (8), US-05.2 Review and Edit AI Plan (5), US-05.3 Approve AI Plan (2)	46	✓
Sprint 2	3–4	US-06.1 Generate Study Plan (8), US-07.1 Auto-Schedule Tasks (8), US-07.4 Mark Task Done (2), US-07.2 Resolve Scheduling Conflicts (5), US-08.1 Detect Falling Behind (3), US-17.1 Today and Deadlines (3), US-17.2 Goal Status (2), US-17.3 Habit Streaks (2), US-18.1 Deadline and Milestone Reminders (5), US-18.2 Task and Habit Reminders (3), US-02.4 Archive Goal (2), US-02.5 Delete Goal (1), US-03.3 Update Course (2), US-03.4 Remove Course (2), US-04.4 Exam Preparation Reminder (2)	50	✓
Sprint 3	5–6	US-06.2 Regenerate Study Plan (3), US-07.3 Adjust Schedule Manually (3), US-13.3 Link Habit to Goal (2), US-14.1 View Goal Workload (5), US-14.2 Overloaded Goal (2), US-08.2 Get Re-planning Recommendations (5), US-08.3 Respond to Recommendations (3), US-15.1 Completion Likelihood (5), US-15.2 Early Warning (5), US-15.3 Recovery Actions (3), US-17.4 Workload Summary (2), US-18.3 Early Warning Notification (2), US-18.4 Streak Milestone Notification (2), US-20.2 Create Student Account (2), US-20.1 View and Update Student Account (3), US-20.3 Suspend Student Account (2)	49	✓
Sprint 4	7–8	US-20.4 Usage Statistics (5), US-20.5 Resolve Account Issues (3), US-09.1 Choose Target Career (2), US-09.2 Generate Career Roadmap (5), US-09.3 Track Roadmap Progress (2), US-09.4 Change Target Career (3), US-10.1 Add Skill (2), US-10.2 View Skill Gaps (5), US-10.3 Update Skill Level (1), US-11.1 Add Project (5), US-11.2 Link Project	49	✓

Sprint	Weeks	User stories	Points	Fits? (≤ 50)
		to Skills (3), US-11.3 View Portfolio (3), US-12.1 Track Applications (3), US-12.2 Interview Preparation (2), US-12.3 Track CV and Portfolio Updates (3), US-12.4 Record Outcome and Feedback (2)		
Sprint 5	9–10	US-16.1 Personalized Study Advice (5), US-16.2 Recovery Strategy (3), US-16.3 Motivational Guidance (2), US-17.5 Career Progress (1), US-19.1 Create Study Group (2), US-19.2 Join Study Group (2), US-19.3 Share and Track Group Progress (5), US-19.4 Accountability Partner (5), US-19.5 Support Messages (3), US-20.6 Retention Metrics (3)	31	✓
Sprint 6	11–12	Buffer for unfinished stories	0	✓

- US-06.1 is next after Sprint 1, but 46 + 8 = 54 > 50 ✗, so it moves to the top of Sprint 2.
- US-06.2 is next after Sprint 2, but 50 + 3 = 53 > 50 ✗, so it moves to the top of Sprint 3.
- US-20.4 is next after Sprint 3, but 49 + 5 = 54 > 50 ✗, so it moves to the top of Sprint 4.
- US-16.1 is next after Sprint 4, but 49 + 5 = 54 > 50 ✗, so it moves to the top of Sprint 5.

Burndown

Point in time	Remaining (ideal)	Remaining (planned)	Remaining (actual)
Start	225	225	225
End of Sprint 1	180	179	—
End of Sprint 2	135	129	—
End of Sprint 3	90	80	—
End of Sprint 4	45	31	—
End of Sprint 5	0	0	—

The ideal line drops by the same amount every sprint (225 ÷ 5 = 45); the planned line follows the sprint plan. The actual column is filled in after each sprint.

C10 — PERT Estimation Error

For every story the team gave three estimates in days: **O** (optimistic), **M** (most likely = points × 1 day, from C9) and **P** (pessimistic). O and P depend on how uncertain the story is: simple screens and forms have P close to M, while stories that use the AI provider, the email service or the scheduling algorithm have a much larger P. P is always farther from M than O, because things go wrong more often than they go better.

```
E  = (O + 4M + P) ÷ 6
SD = (P − O) ÷ 6
```

User story	O (days)	M (days)	P (days)	E = (O+4M+P)/6	SD = (P−O)/6
US-01.1 Create Profile	2	3	5	19 ÷ 6 = 3.17	3 ÷ 6 = 0.50
US-01.2 Update Profile	1.5	2	3	12.5 ÷ 6 = 2.08	1.5 ÷ 6 = 0.25
US-01.3 View Profile	0.5	1	2	6.5 ÷ 6 = 1.08	1.5 ÷ 6 = 0.25
US-02.1 Create Goal	2	3	5	19 ÷ 6 = 3.17	3 ÷ 6 = 0.50
US-02.2 Edit Goal	1.5	2	3	12.5 ÷ 6 = 2.08	1.5 ÷ 6 = 0.25
US-02.3 Complete Goal	0.5	1	2	6.5 ÷ 6 = 1.08	1.5 ÷ 6 = 0.25
US-02.4 Archive Goal	1.5	2	3	12.5 ÷ 6 = 2.08	1.5 ÷ 6 = 0.25
US-02.5 Delete Goal	0.5	1	2	6.5 ÷ 6 = 1.08	1.5 ÷ 6 = 0.25
US-03.1 Add Course	2	3	5	19 ÷ 6 = 3.17	3 ÷ 6 = 0.50
US-03.2 View Course Progress	2	3	6	20 ÷ 6 = 3.33	4 ÷ 6 = 0.67
US-03.3 Update Course	1.5	2	3	12.5 ÷ 6 = 2.08	1.5 ÷ 6 = 0.25
US-03.4 Remove Course	1.5	2	3	12.5 ÷ 6 = 2.08	1.5 ÷ 6 = 0.25
US-04.1 Add Assignment	2	3	5	19 ÷ 6 = 3.17	3 ÷ 6 = 0.50
US-04.2 Add Exam	1.5	2	3	12.5 ÷ 6 = 2.08	1.5 ÷ 6 = 0.25
US-04.3 Track Assignment Status	1.5	2	3	12.5 ÷ 6 = 2.08	1.5 ÷ 6 = 0.25
US-04.4 Exam Preparation Reminder	1.5	2	3	12.5 ÷ 6 = 2.08	1.5 ÷ 6 = 0.25
US-05.1 Generate Goal Plan	6	8	18	56 ÷ 6 = 9.33	12 ÷ 6 = 2.00
US-05.2 Review and Edit AI Plan	4	5	10	34 ÷ 6 = 5.67	6 ÷ 6 = 1.00

User story	O (days)	M (days)	P (days)	E = (O+4M+P)/6	SD = (P−O)/6
US-05.3 Approve AI Plan	1.5	2	3	12.5 ÷ 6 = 2.08	1.5 ÷ 6 = 0.25
US-06.1 Generate Study Plan	6	8	18	56 ÷ 6 = 9.33	12 ÷ 6 = 2.00
US-06.2 Regenerate Study Plan	2	3	8	22 ÷ 6 = 3.67	6 ÷ 6 = 1.00
US-07.1 Auto-Schedule Tasks	6	8	18	56 ÷ 6 = 9.33	12 ÷ 6 = 2.00
US-07.2 Resolve Scheduling Conflicts	4	5	10	34 ÷ 6 = 5.67	6 ÷ 6 = 1.00
US-07.3 Adjust Schedule Manually	2	3	6	20 ÷ 6 = 3.33	4 ÷ 6 = 0.67
US-07.4 Mark Task Done	1.5	2	3	12.5 ÷ 6 = 2.08	1.5 ÷ 6 = 0.25
US-08.1 Detect Falling Behind	2	3	6	20 ÷ 6 = 3.33	4 ÷ 6 = 0.67
US-08.2 Get Re-planning Recommendations	4	5	10	34 ÷ 6 = 5.67	6 ÷ 6 = 1.00
US-08.3 Respond to Recommendations	2	3	6	20 ÷ 6 = 3.33	4 ÷ 6 = 0.67
US-09.1 Choose Target Career	1.5	2	3	12.5 ÷ 6 = 2.08	1.5 ÷ 6 = 0.25
US-09.2 Generate Career Roadmap	4	5	12	36 ÷ 6 = 6.00	8 ÷ 6 = 1.33
US-09.3 Track Roadmap Progress	1.5	2	3	12.5 ÷ 6 = 2.08	1.5 ÷ 6 = 0.25
US-09.4 Change Target Career	2	3	8	22 ÷ 6 = 3.67	6 ÷ 6 = 1.00
US-10.1 Add Skill	1.5	2	3	12.5 ÷ 6 = 2.08	1.5 ÷ 6 = 0.25
US-10.2 View Skill Gaps	4	5	10	34 ÷ 6 = 5.67	6 ÷ 6 = 1.00
US-10.3 Update Skill Level	0.5	1	2	6.5 ÷ 6 = 1.08	1.5 ÷ 6 = 0.25
US-11.1 Add Project	4	5	10	34 ÷ 6 = 5.67	6 ÷ 6 = 1.00
US-11.2 Link Project to Skills	2	3	6	20 ÷ 6 = 3.33	4 ÷ 6 = 0.67
US-11.3 View Portfolio	2	3	5	19 ÷ 6 = 3.17	3 ÷ 6 = 0.50
US-12.1 Track Applications	2	3	5	19 ÷ 6 = 3.17	3 ÷ 6 = 0.50

User story	O (days)	M (days)	P (days)	E = (O+4M+P)/6	SD = (P−O)/6
US-12.2 Interview Preparation	1.5	2	3	12.5 ÷ 6 = 2.08	1.5 ÷ 6 = 0.25
US-12.3 Track CV and Portfolio Updates	2	3	5	19 ÷ 6 = 3.17	3 ÷ 6 = 0.50
US-12.4 Record Outcome and Feedback	1.5	2	3	12.5 ÷ 6 = 2.08	1.5 ÷ 6 = 0.25
US-13.1 Create Habit	2	3	5	19 ÷ 6 = 3.17	3 ÷ 6 = 0.50
US-13.2 Complete Habit and Keep Streak	2	3	6	20 ÷ 6 = 3.33	4 ÷ 6 = 0.67
US-13.3 Link Habit to Goal	1.5	2	3	12.5 ÷ 6 = 2.08	1.5 ÷ 6 = 0.25
US-14.1 View Goal Workload	4	5	10	34 ÷ 6 = 5.67	6 ÷ 6 = 1.00
US-14.2 Overloaded Goal	1.5	2	3	12.5 ÷ 6 = 2.08	1.5 ÷ 6 = 0.25
US-15.1 Completion Likelihood	4	5	10	34 ÷ 6 = 5.67	6 ÷ 6 = 1.00
US-15.2 Early Warning	4	5	10	34 ÷ 6 = 5.67	6 ÷ 6 = 1.00
US-15.3 Recovery Actions	2	3	6	20 ÷ 6 = 3.33	4 ÷ 6 = 0.67
US-16.1 Personalized Study Advice	4	5	12	36 ÷ 6 = 6.00	8 ÷ 6 = 1.33
US-16.2 Recovery Strategy	2	3	8	22 ÷ 6 = 3.67	6 ÷ 6 = 1.00
US-16.3 Motivational Guidance	1.5	2	3	12.5 ÷ 6 = 2.08	1.5 ÷ 6 = 0.25
US-17.1 Today and Deadlines	2	3	5	19 ÷ 6 = 3.17	3 ÷ 6 = 0.50
US-17.2 Goal Status	1.5	2	3	12.5 ÷ 6 = 2.08	1.5 ÷ 6 = 0.25
US-17.3 Habit Streaks	1.5	2	3	12.5 ÷ 6 = 2.08	1.5 ÷ 6 = 0.25
US-17.4 Workload Summary	1.5	2	3	12.5 ÷ 6 = 2.08	1.5 ÷ 6 = 0.25
US-17.5 Career Progress	0.5	1	2	6.5 ÷ 6 = 1.08	1.5 ÷ 6 = 0.25
US-18.1 Deadline and Milestone Reminders	4	5	12	36 ÷ 6 = 6.00	8 ÷ 6 = 1.33
US-18.2 Task and Habit Reminders	2	3	6	20 ÷ 6 = 3.33	4 ÷ 6 = 0.67

User story	O (days)	M (days)	P (days)	E = (O+4M+P)/6	SD = (P−O)/6
US-18.3 Early Warning Notification	1.5	2	3	12.5 ÷ 6 = 2.08	1.5 ÷ 6 = 0.25
US-18.4 Streak Milestone Notification	1.5	2	3	12.5 ÷ 6 = 2.08	1.5 ÷ 6 = 0.25
US-19.1 Create Study Group	1.5	2	3	12.5 ÷ 6 = 2.08	1.5 ÷ 6 = 0.25
US-19.2 Join Study Group	1.5	2	3	12.5 ÷ 6 = 2.08	1.5 ÷ 6 = 0.25
US-19.3 Share and Track Group Progress	4	5	10	34 ÷ 6 = 5.67	6 ÷ 6 = 1.00
US-19.4 Accountability Partner	4	5	10	34 ÷ 6 = 5.67	6 ÷ 6 = 1.00
US-19.5 Support Messages	2	3	5	19 ÷ 6 = 3.17	3 ÷ 6 = 0.50
US-20.1 View and Update Student Account	2	3	5	19 ÷ 6 = 3.17	3 ÷ 6 = 0.50
US-20.2 Create Student Account	1.5	2	3	12.5 ÷ 6 = 2.08	1.5 ÷ 6 = 0.25
US-20.3 Suspend Student Account	1.5	2	3	12.5 ÷ 6 = 2.08	1.5 ÷ 6 = 0.25
US-20.4 Usage Statistics	4	5	10	34 ÷ 6 = 5.67	6 ÷ 6 = 1.00
US-20.5 Resolve Account Issues	2	3	5	19 ÷ 6 = 3.17	3 ÷ 6 = 0.50
US-20.6 Retention Metrics	2	3	6	20 ÷ 6 = 3.33	4 ÷ 6 = 0.67
Total		**225**		**249.00**	

Step 1 — Estimation error of one story

```
US-05.1 Generate Goal Plan: E = 9.33 days, SD = 2.00
Range (68%) = 9.33 ± 2.00 → from about 7.33 to 11.33 days
```

US-05.1 Generate Goal Plan, US-06.1 Generate Study Plan, US-07.1 Auto-Schedule Tasks have the largest SD (2.00) → they are the riskiest stories; the team should watch them closely.

Step 2 — Total expected effort

```
E(total) = Σ (O + 4M + P) ÷ 6 = 1494 ÷ 6 = 249.00 person-days
Single-point estimate (ΣM) = 225 person-days
```

```
Difference = 249.00 − 225 = 24.00 days
Percentage = 24.00 ÷ 225 × 100 = 10.7%
```

The single-point estimate was **24.00 days (10.7%) optimistic**: it planned for less work than the team will most probably need.

The total is computed from unrounded per-story values. The per-story rows are rounded to two decimals, so summing the displayed rows gives 248.94 rather than 249.00.

Step 3 — Is the team still enough?

```
Effort (person-weeks) = 249.00 ÷ 5 = 49.80 person-weeks
Team size = 49.80 ÷ 12 = 4.15 → round up → 5 developers
Capacity = 5 × 12 = 60 person-weeks ≥ 49.80 ✓
Number of sprints = 249.00 ÷ 50 = 4.98 → 5 sprints ≤ 6 ✓
```

Yes, the team is still enough. Our 5 developers can still finish before the deadline, but with a smaller safety margin (60 vs. 49.80 instead of 60 vs. 45), and the team needed is now exactly our 5 developers. Sprint 6 is kept free as a buffer for overruns.

C11 — Jira Board

Jira details

Field	Value
Jira site	https://aspirra.atlassian.net
Project name	Aspirra Project
Project key	SCRUM
Board	SCRUM board (Board ID: 1)
Issue types used	Epic, Task (stories created as Task with `story` label; no native `Story` issue type available)
Team-managed	Yes

Epics

- Created **20 epics** (one per high-level functional requirement FR-01–FR-20), linked via parent relationship.
- Epic keys: **SCRUM-10** (FR-01) through **SCRUM-27** (FR-20).
- Each epic includes owner, service, linked customer requirements, and labels `epic`, `FR-XX`, and the corresponding service label.

Product backlog (user stories)

- Created **73 user stories** as Jira Tasks (`Task` issue type) with label `story`.

- Stories are linked to their parent epics (e.g., US-01.1–US-01.3 under FR-01 / SCRUM-10).
- Each story includes the user story text, acceptance criteria, traceability (covered low-level FRs), priority, and service labels.
- **Total story points: 225**.
- **Assignees:** None (stories remain unassigned as requested).
- Story keys range from **SCRUM-101** (US-02.1) to **SCRUM-173** (US-20.6).

Sprint 1

Field	Value
Sprint ID	1
Sprint name	Sprint 1 — Foundation
State	Active
Start date	2026-10-22
End date	2026-11-04
Goal	Deliver the core academic foundation, profile, goals, courses, assessments, habits, and the first AI goal planner.
Stories committed	16
Points committed	46

Sprint 1 stories (by key): SCRUM-104 (US-01.1, 3), SCRUM-102 (US-01.3, 1), SCRUM-103 (US-01.2, 2), SCRUM-101 (US-02.1, 3), SCRUM-105 (US-02.2, 2), SCRUM-106 (US-02.3, 1), SCRUM-108 (US-03.1, 3), SCRUM-113 (US-04.1, 3), SCRUM-114 (US-04.2, 2), SCRUM-115 (US-04.3, 2), SCRUM-109 (US-03.2, 3), SCRUM-143 (US-13.1, 3), SCRUM-144 (US-13.2, 3), SCRUM-117 (US-05.1, 8), SCRUM-118 (US-05.2, 5), SCRUM-119 (US-05.3, 2).

Reports

- **Sprint report / burndown:** SCRUM Sprint 1 — burndown/report

Cleanup performed

- Deleted sample issues **SCRUM-1**–**SCRUM-4**.
- Deleted **SCRUM Sprint 0**.
- Renamed/reconfigured the original future sprint to **Sprint 1 — Foundation** with dates 2026-10-22–2026-11-04, goal set, and state set to **Active**.

C12 — GitHub Repository

- **Repository:** https://github.com/AIU-SoftWave/Aspirra
- **Branches:** `main` (protected), `develop`, `feature/<service>` per owner
- **Collaborators:** all five team members (TA invitation pending the TA's account details)
- **README:** title/description, team, services and owners, tech stack, folder structure, Jira link

Part D — GUI Design

Figma GUI screens covering the system functionality.

D13 — Figma GUI Screens

- **Design source:** Figma; the exported screens are stored in `docs/proposal/gui/`.

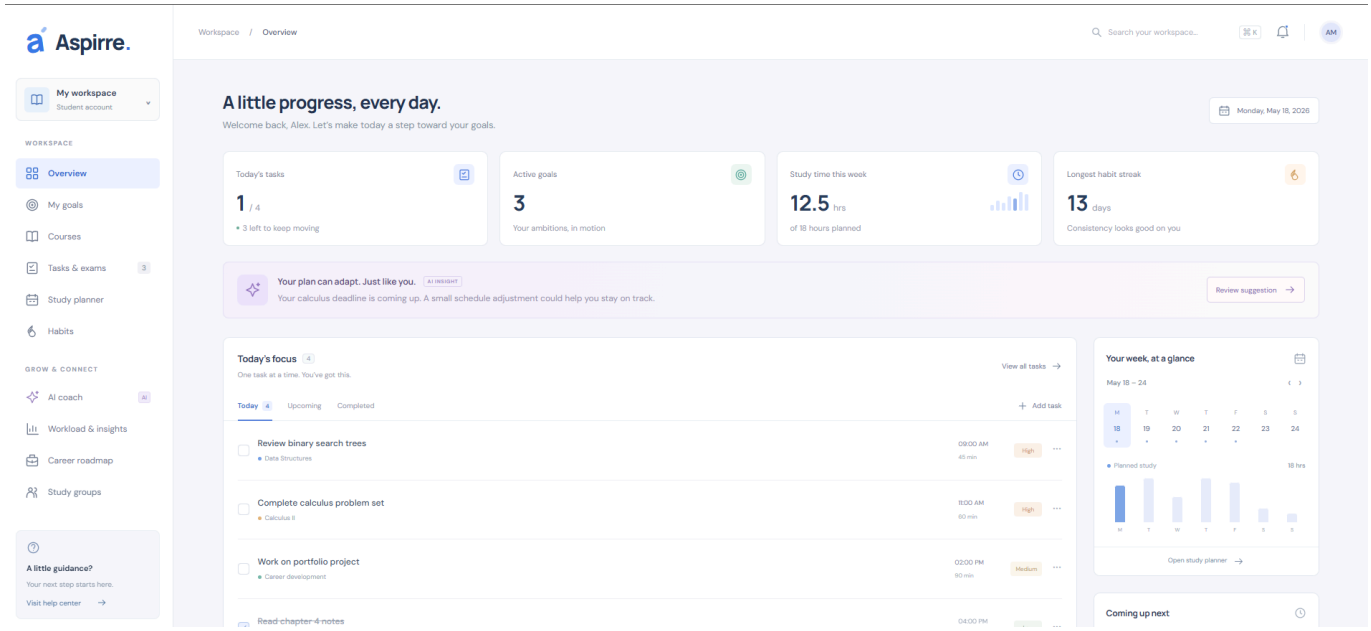
Screens

The mockups below cover the primary student experience of Aspirra.

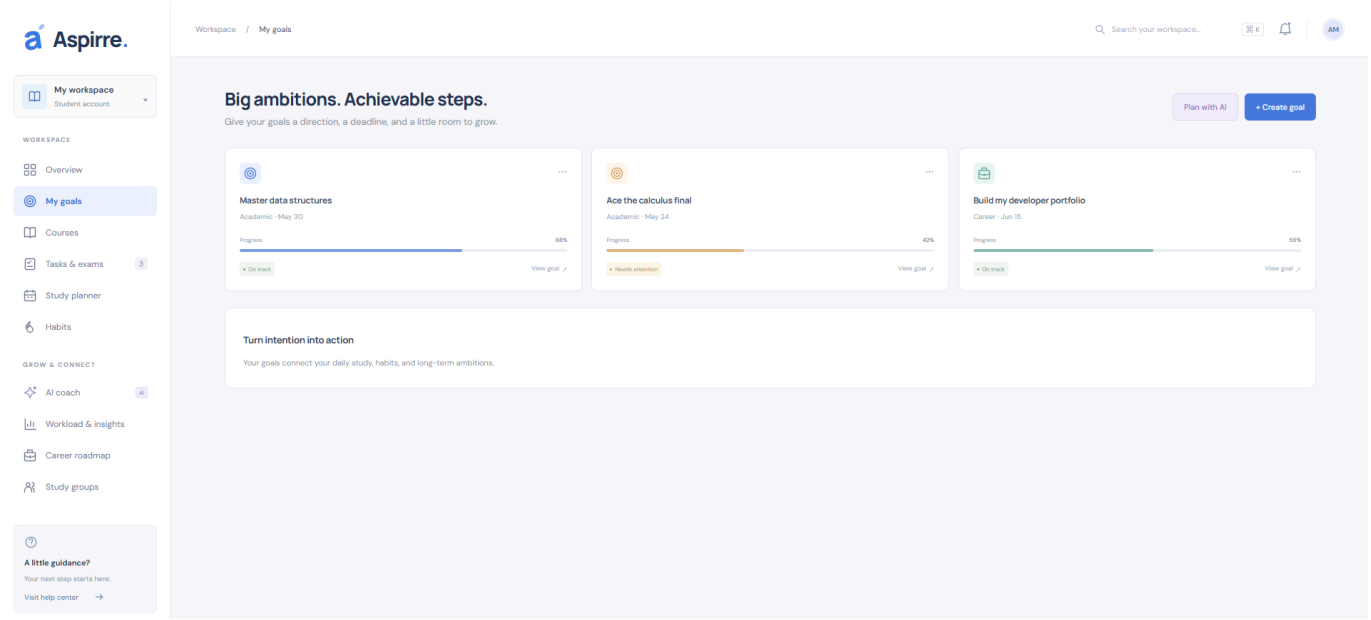
#	Screen	Covers functionality
1	Dashboard overview	FR-17 Display Student Dashboard
2	Goals	FR-02 Manage Academic Goals, FR-05 Plan Goals with AI
3	Courses	FR-03 Manage Courses
4	Tasks & Exams	FR-04 Manage Assignments and Exams
5	Study Planner	FR-06 Generate Study Plans, FR-07 Enable Smart Scheduling, FR-08 Enable Adaptive Planning
6	Habits	FR-13 Manage Habits
7	AI Coach	FR-16 Coach Students with AI
8	Workload & Insights	FR-14 Analyze Study Workload, FR-15 Predict and Warn Early
9	Career Roadmap	FR-09 Manage Career Roadmaps, FR-10 Track Skills, FR-11 Manage Projects and Portfolios, FR-12 Manage Internship and Job Preparation
10	Study Groups	FR-19 Manage Study Groups and Accountability
11	Profile	FR-01 Manage Student Profile

The administration and notification screens are specified in B4 (FR-18 and FR-20) and are not part of this design export.

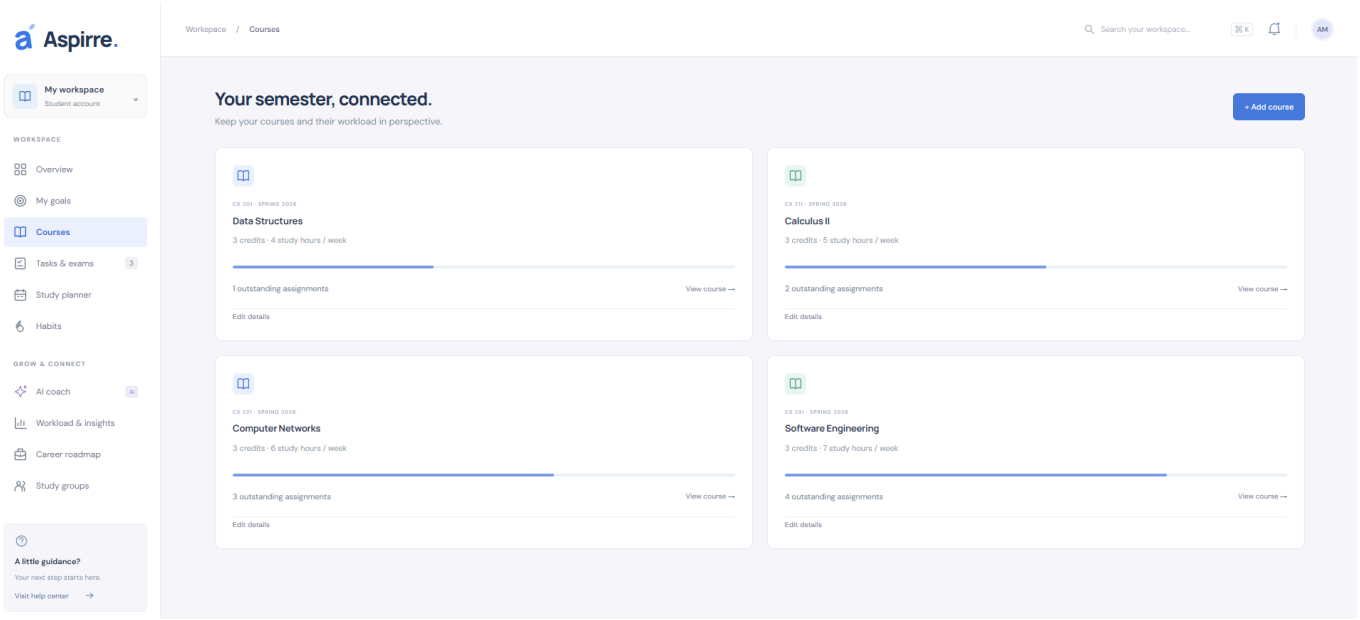
1. Dashboard overview



2. Goals



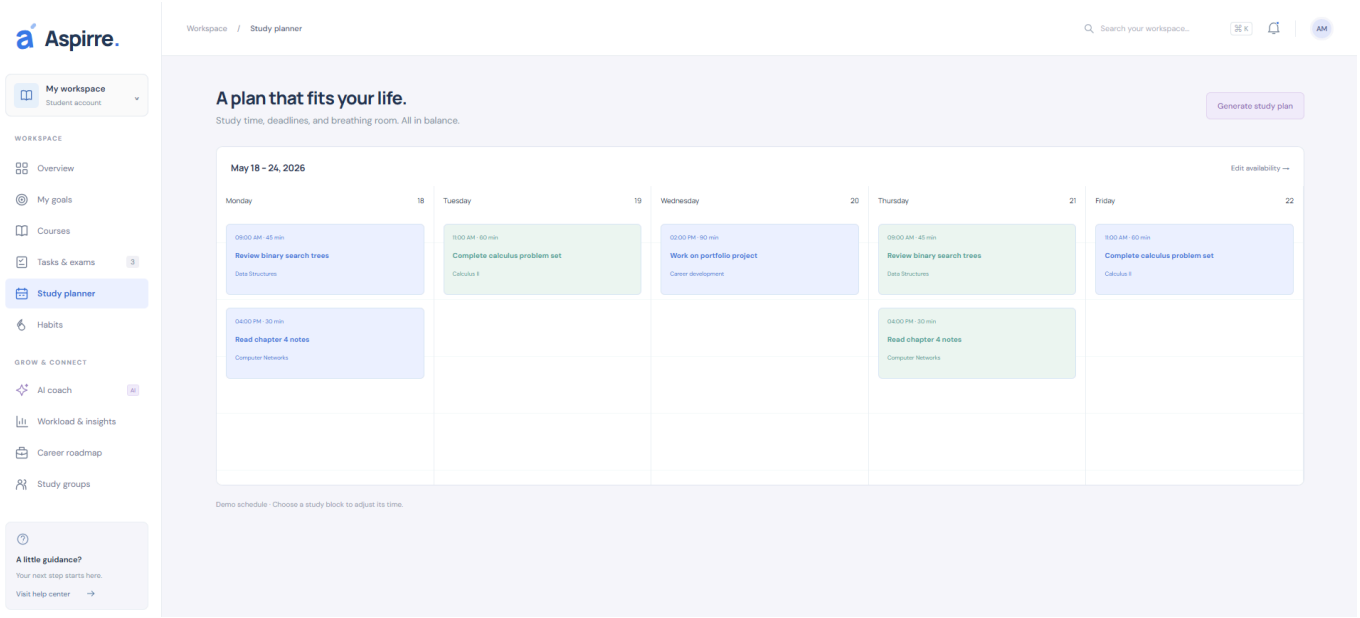
3. Courses



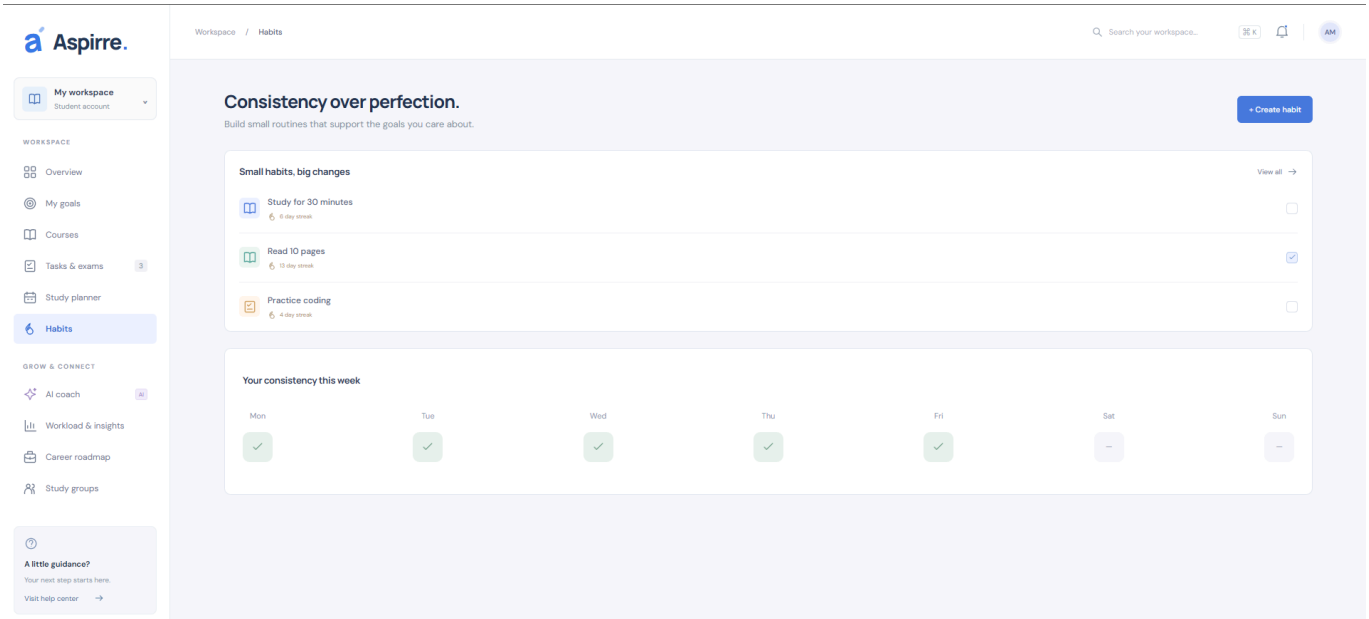
4. Tasks & Exams



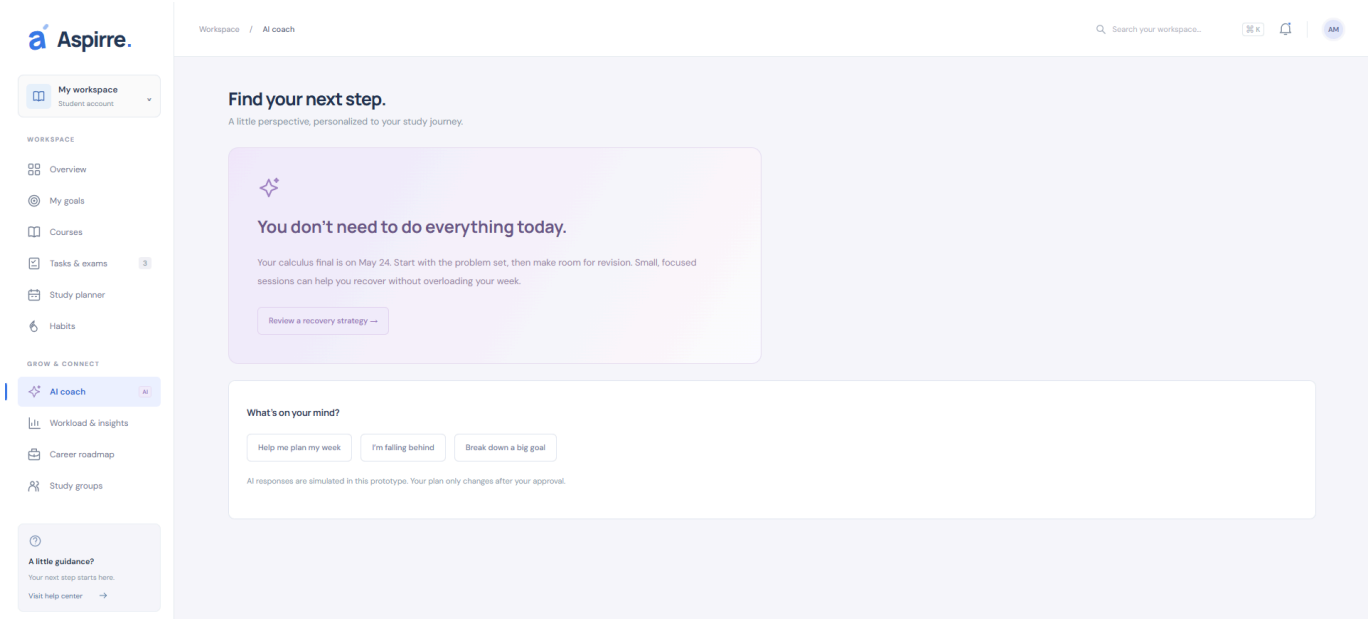
5. Study Planner



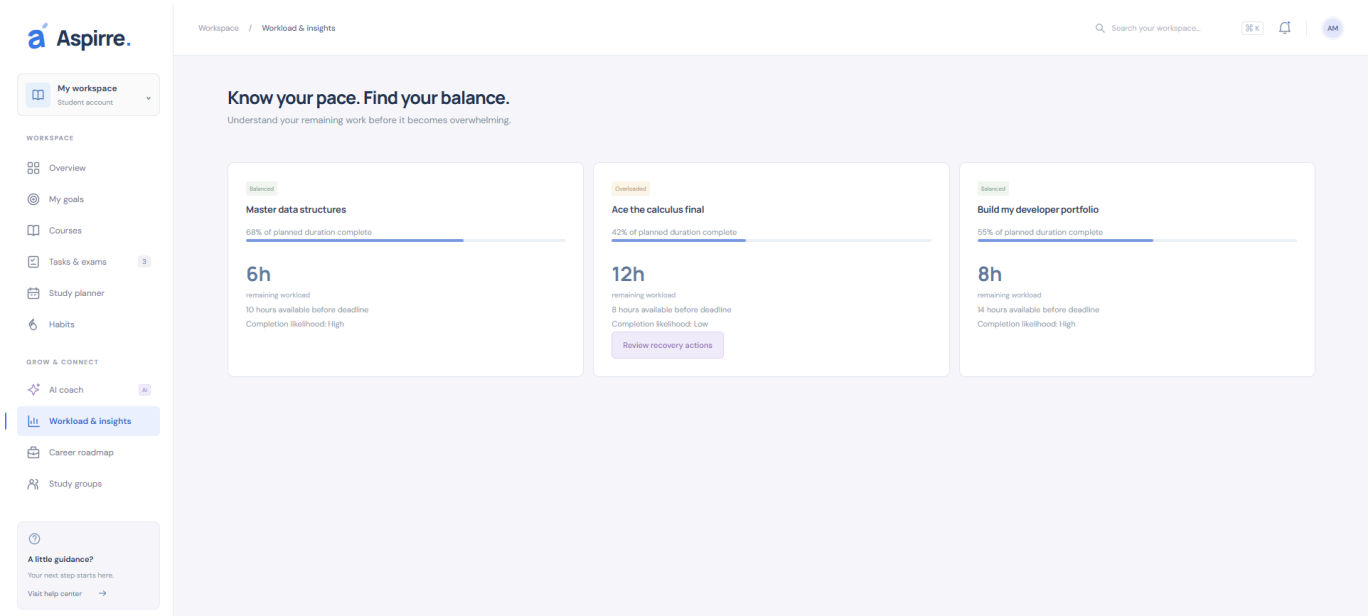
6. Habits



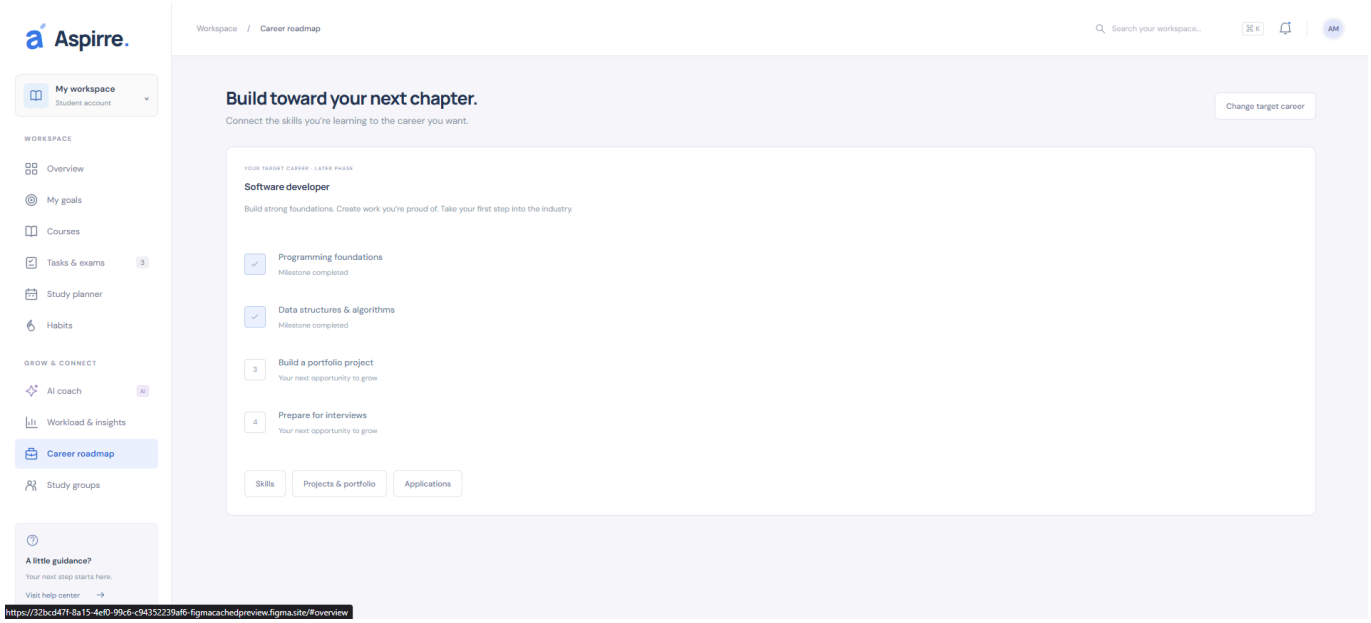
7. AI Coach



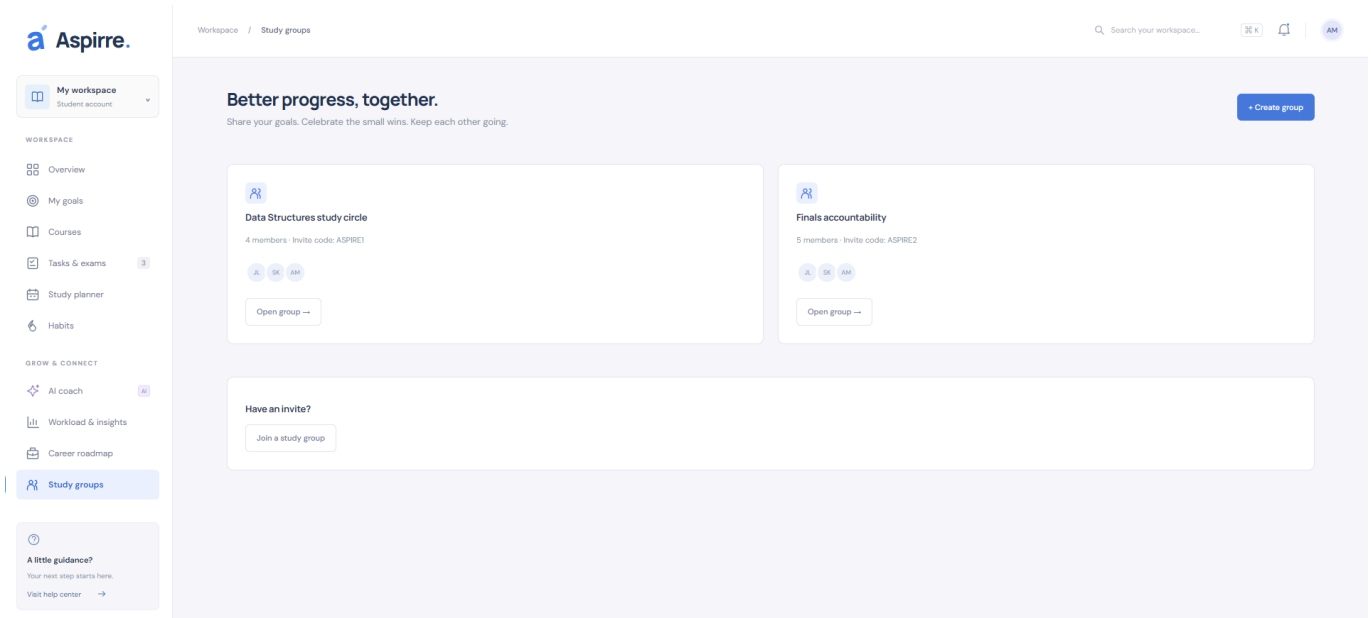
8. Workload & Insights



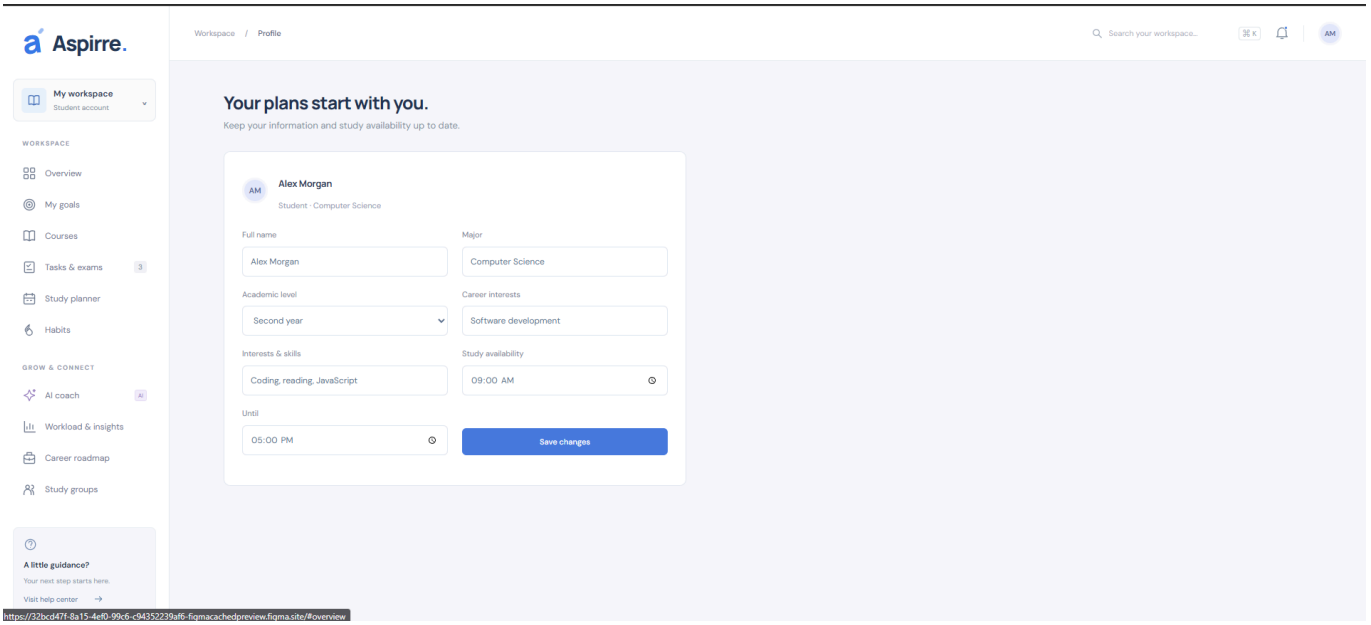
9. Career Roadmap



10. Study Groups



11. Profile


Aspirre.
My workspace
Student account
WORKSPACE
Overview
My goals
Courses
Tasks & exams
3
Study planner
Habits
GROW & CONNECT
AI coach
Workload & insights
Career roadmap
Study groups
A little guidance?
Your next step starts here.
Visit help center
Workspace / Profile
Search your workspace...
AM
Your plans start with you.
Keep your information and study availability up to date.
AM
Alex Morgan
Student · Computer Science
Full name
Alex Morgan
Major
Computer Science
Academic level
Second year
Career interests
Software development
Interests & skills
Coding, reading, JavaScript
Study availability
09:00 AM
Until
05:00 PM
Save changes
https://32bcd47f-8a15-4ef0-99c6-c94352239af6-figmacachedpreview.figma.site/#overview